CIP-A106 Laboratory 4

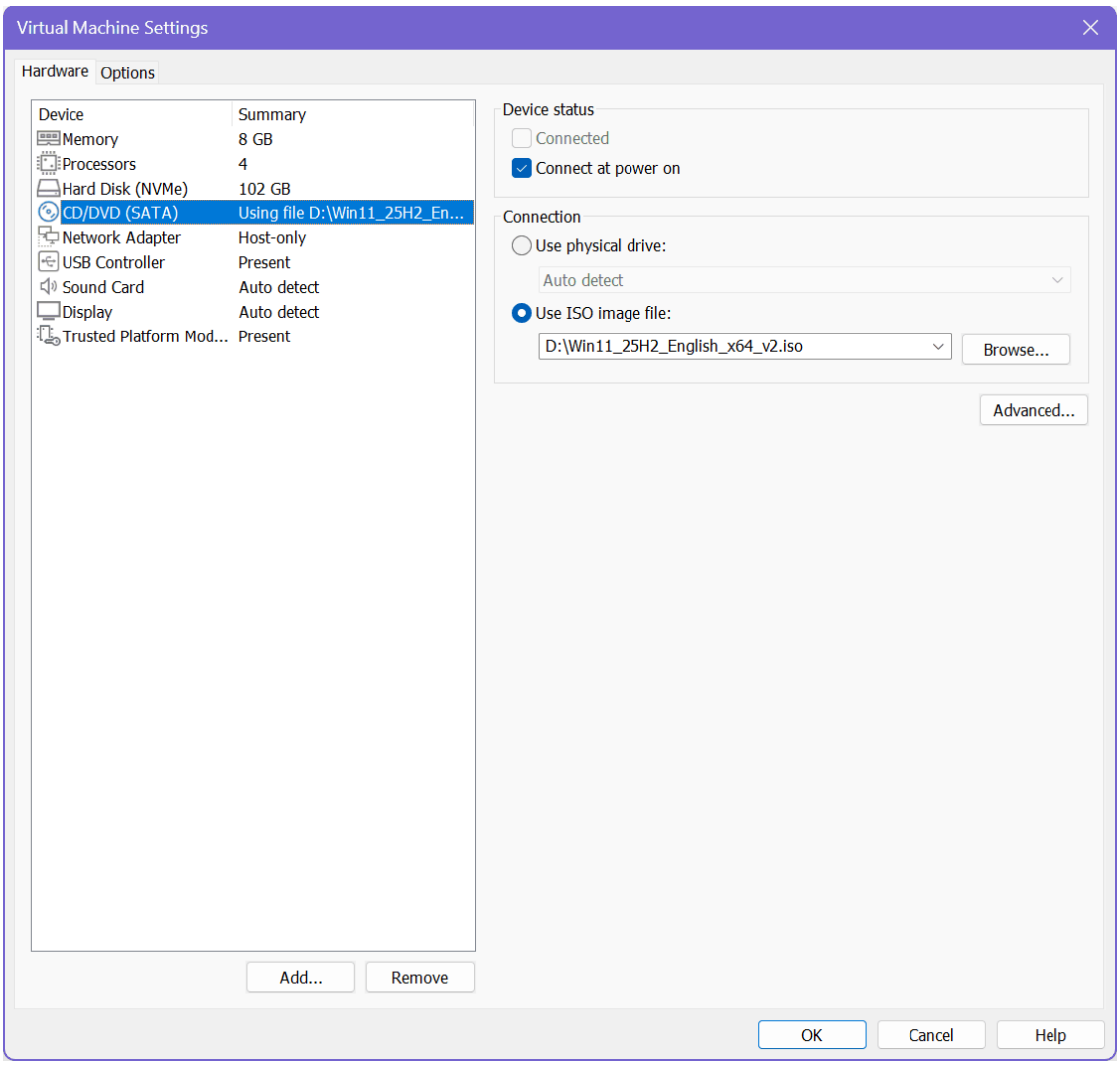
# Evidence Appendix

Screenshots and exports supporting the Technical Report

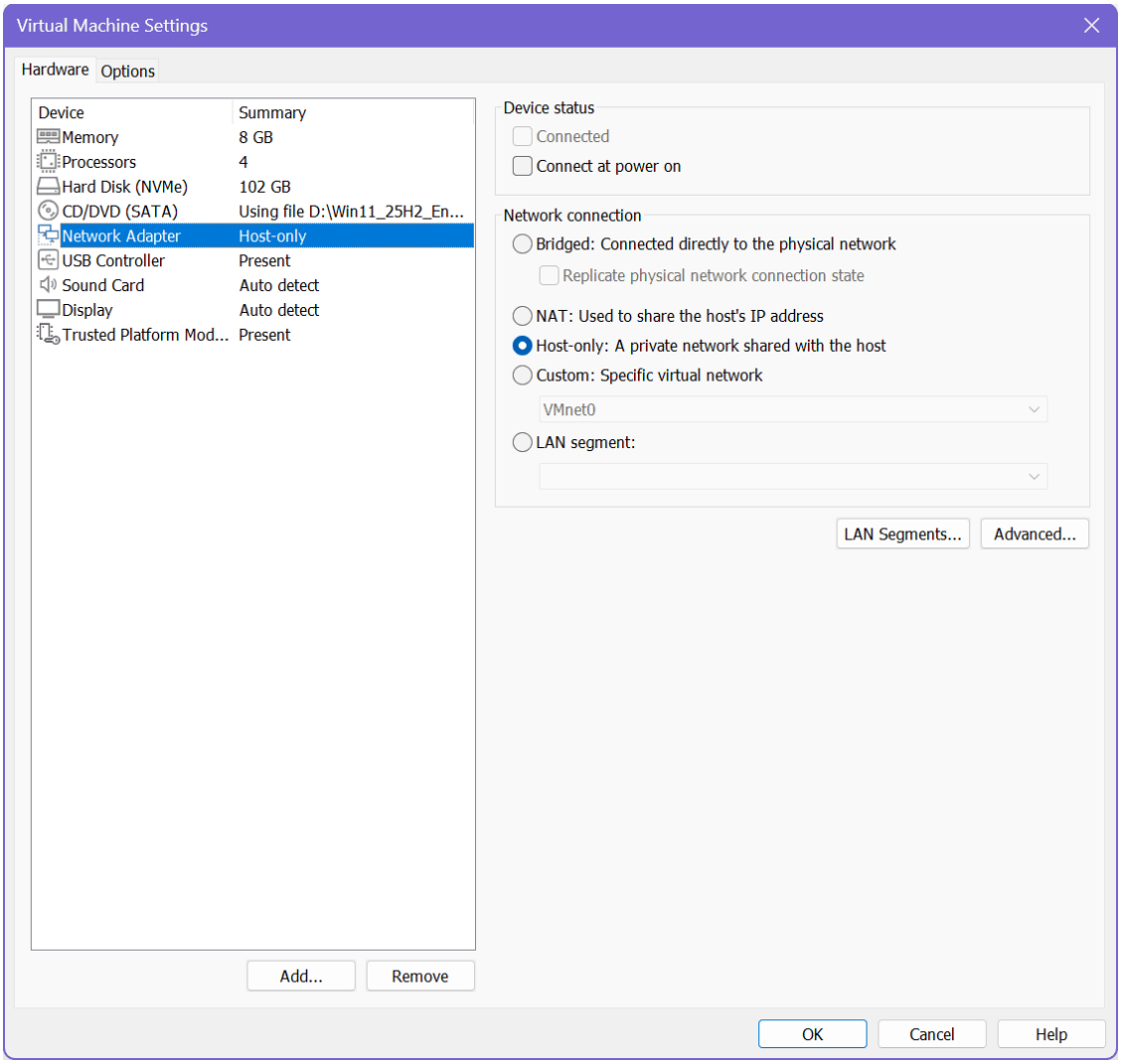
| | |
|---|---|
| **Student** | Perry Koko, C1126EHIT17343 |
| **Contents** | A isolation, snapshots, transfer; B static identification (PEStudio); C Ghidra; D debugger runs; E memory and script context; T rebuilt text exports. Exhibits E01 to E53 follow capture order and match the file names in 05_Screenshots. |
| **Capture method** | Screenshots were taken on the host (the VM has no capture channel). The time in each caption is the clock visible in the screenshot (VM local time). A time marked with * is the capture time of the file, because no clock is visible. Each file has a SHA-256 in SHA256SUMS.txt and in screenshots_manifest.csv. |
| **Excluded** | The sample, the decompiled script, raw memory dumps and the ISO are not included. Domains and URLs shown are text from decoded strings and are not links. |

## A. Isolation, snapshots and transfer (Task A)

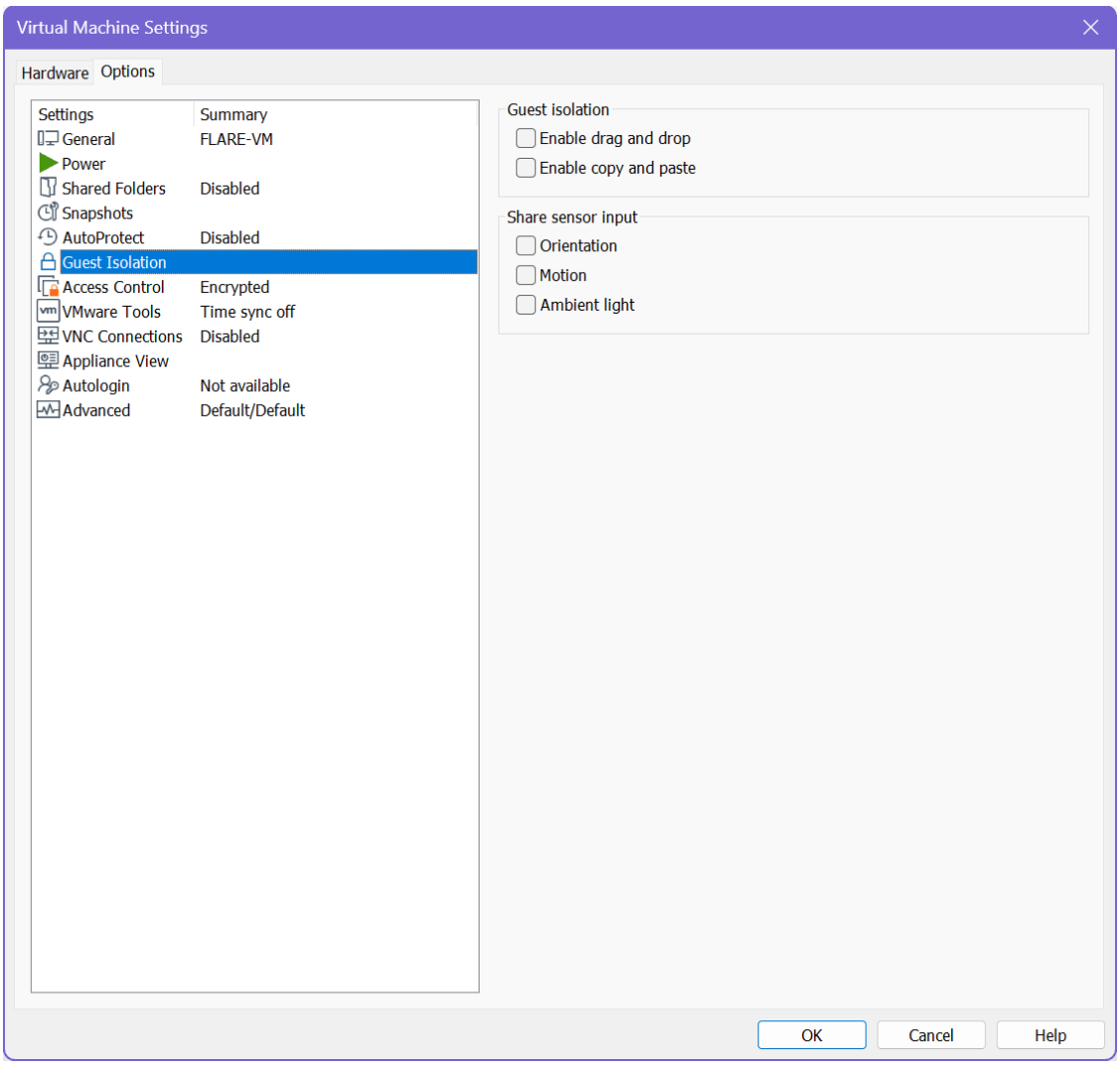
Host-only adapter with both connection boxes unchecked, Shared Folders and VNC disabled, guest isolation unchecked, the two snapshots, and the CD/DVD transfer window.



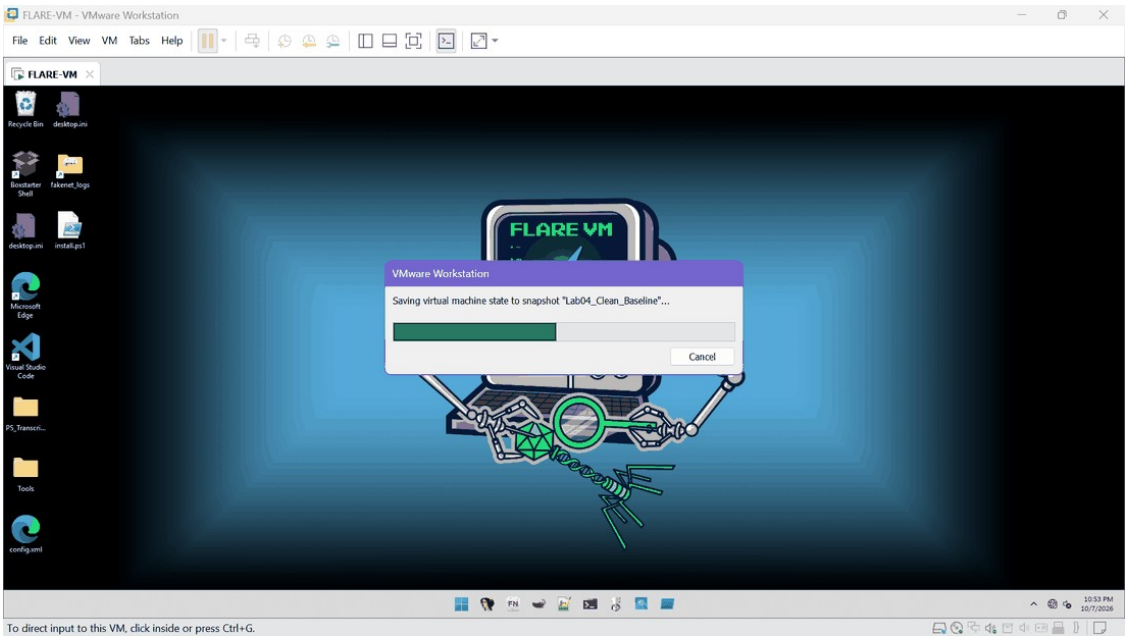

***Ex. E01*** | *2026-10-07 22:56** | *VMware Workstation* | *VM settings, CD/DVD drive as found: Windows 11 installation ISO attached (Connected unchecked, Connect at power on checked)* | *SHA-256 331C600223B8F78F…*



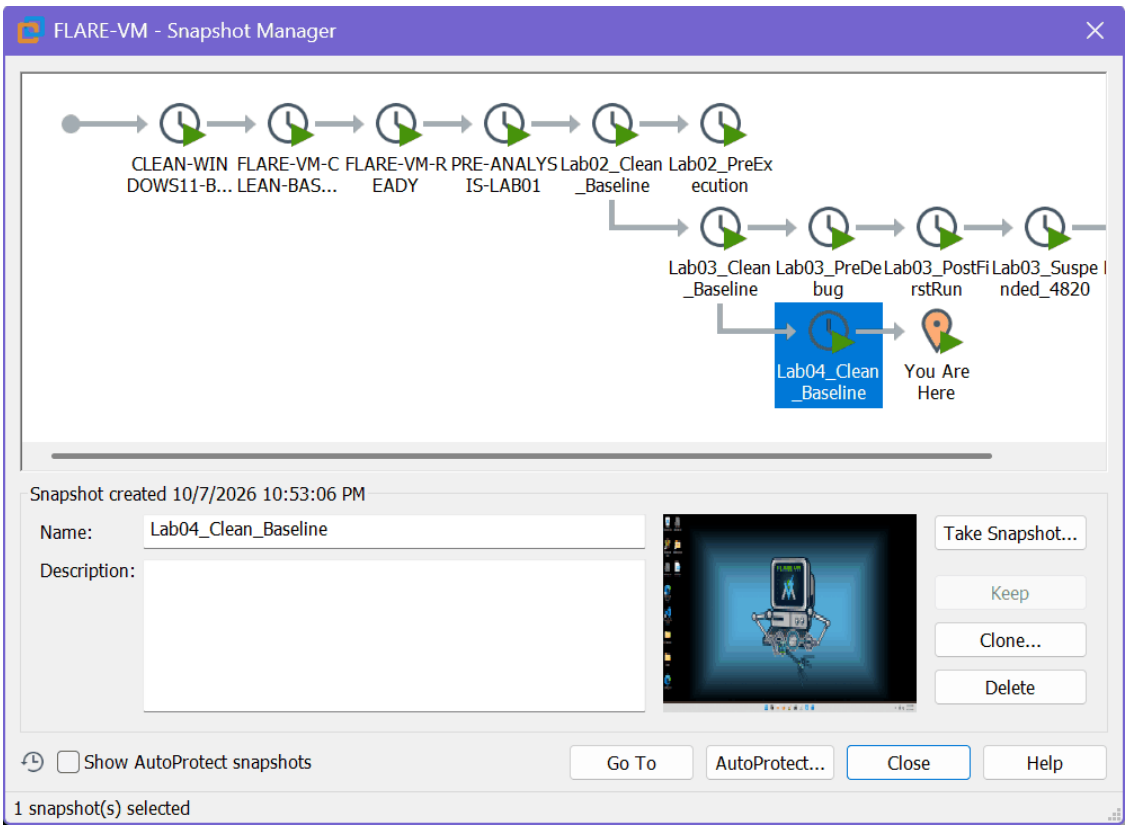

***Ex. E02*** | *2026-10-07 22:56* | VMware Workstation | VM settings, Network Adapter: Host-only; Connected and Connect at power on both unchecked; Bridged, NAT and Custom not selected | SHA-256 83F1DC9115CF9C6C…*



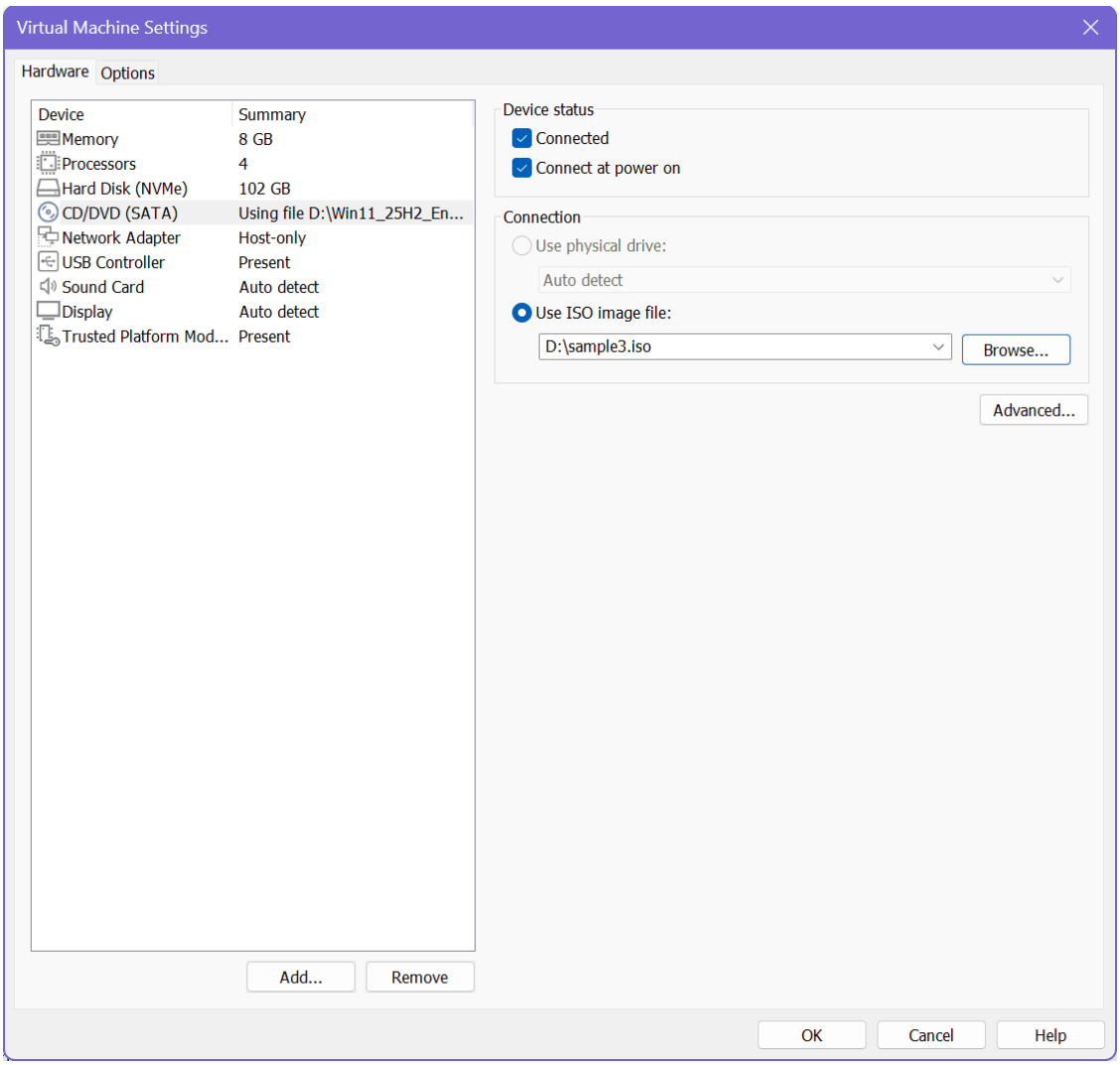

***Ex. E03*** | *2026-10-07 22:56* | VMware Workstation | VM Options: Shared Folders Disabled, VNC Connections Disabled, Guest Isolation with drag and drop and copy and paste both unchecked | SHA-256 2929EF808D67914B…*



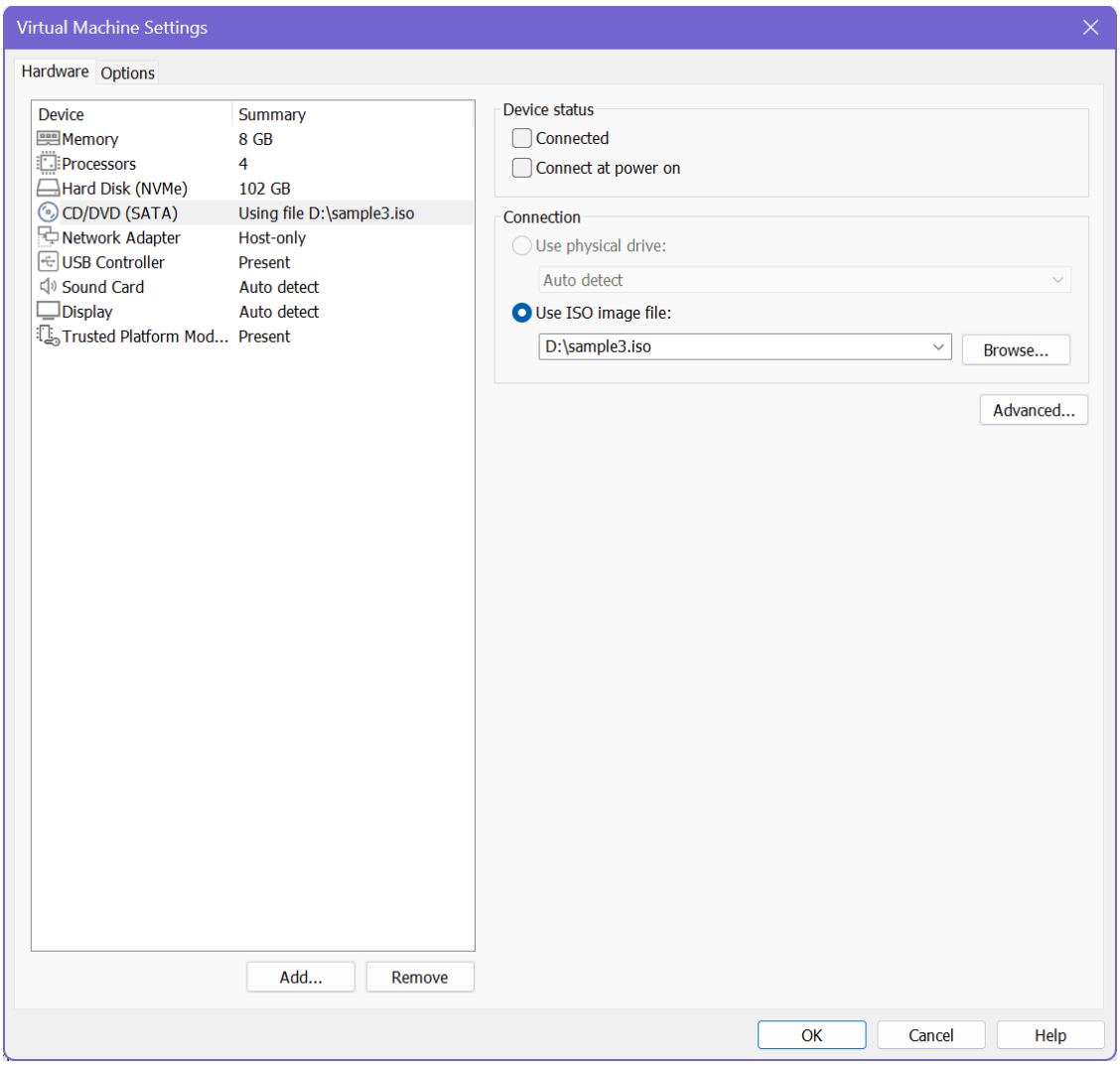

***Ex. E04*** | *2026-10-07 22:53 | VMware Workstation | Saving virtual machine state to snapshot "Lab04_Clean_Baseline" (clock 10:53 PM, 10/7/2026) | SHA-256 990E5B8B715ACE4F…*



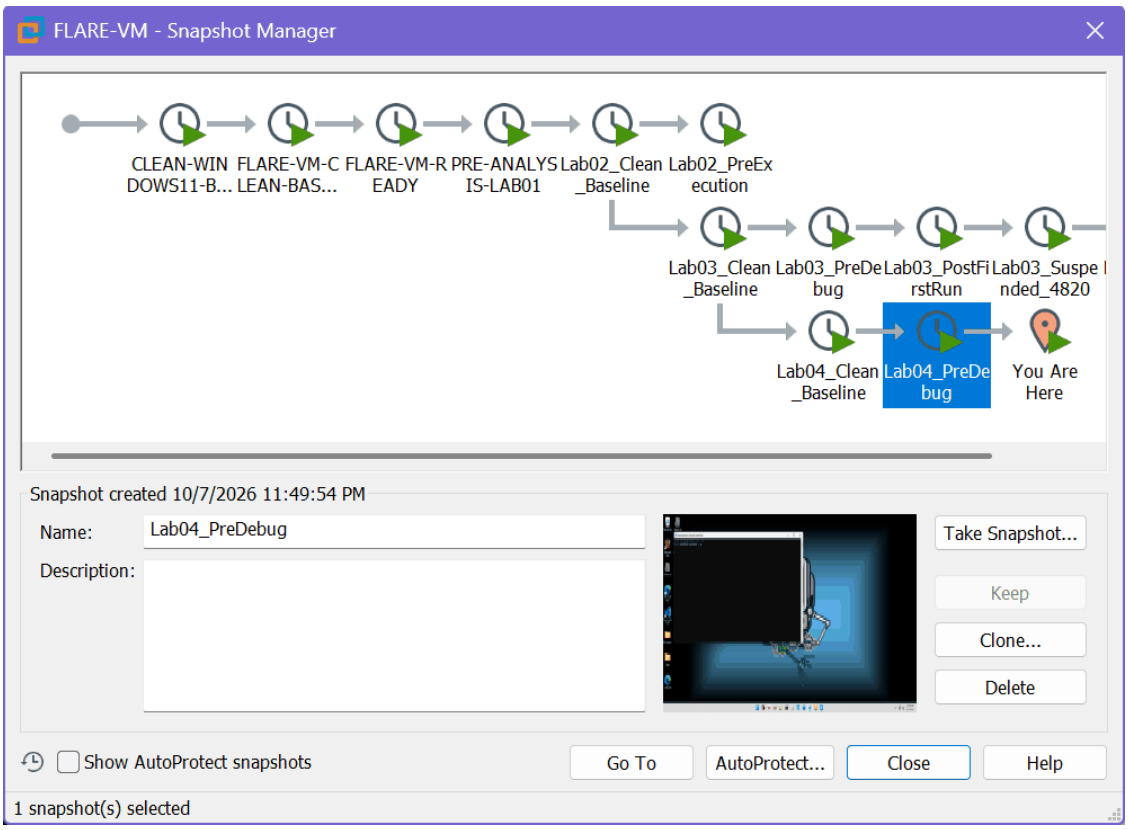

***Ex. E05*** | *2026-10-07 22:53:06 | VMware Workstation | Snapshot Manager: Lab04_Clean_Baseline created 10/7/2026 10:53:06 PM, child of Lab03_Clean_Baseline; You Are Here after it | SHA-256 687B7F51236CAAED...*




***Ex. E06*** | *2026-10-07 23:14** | *VMware Workstation* | *CD/DVD drive set to D:\sample3.iso with Connected checked (one-way transfer window opened)* | *SHA-256 C3408A26284BE6CA…*



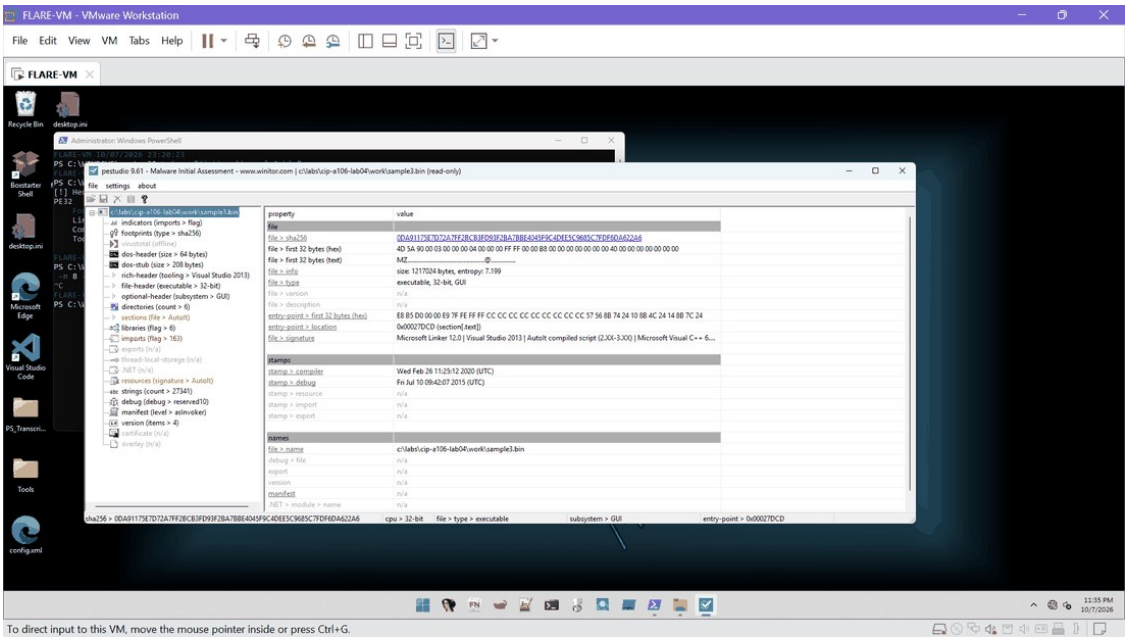

***Ex. E07*** | *2026-10-07 23:17** | *VMware Workstation* | *CD/DVD drive Connected unchecked again after the transfer (transfer window closed)* | *SHA-256 981A58E0B76D0402…*



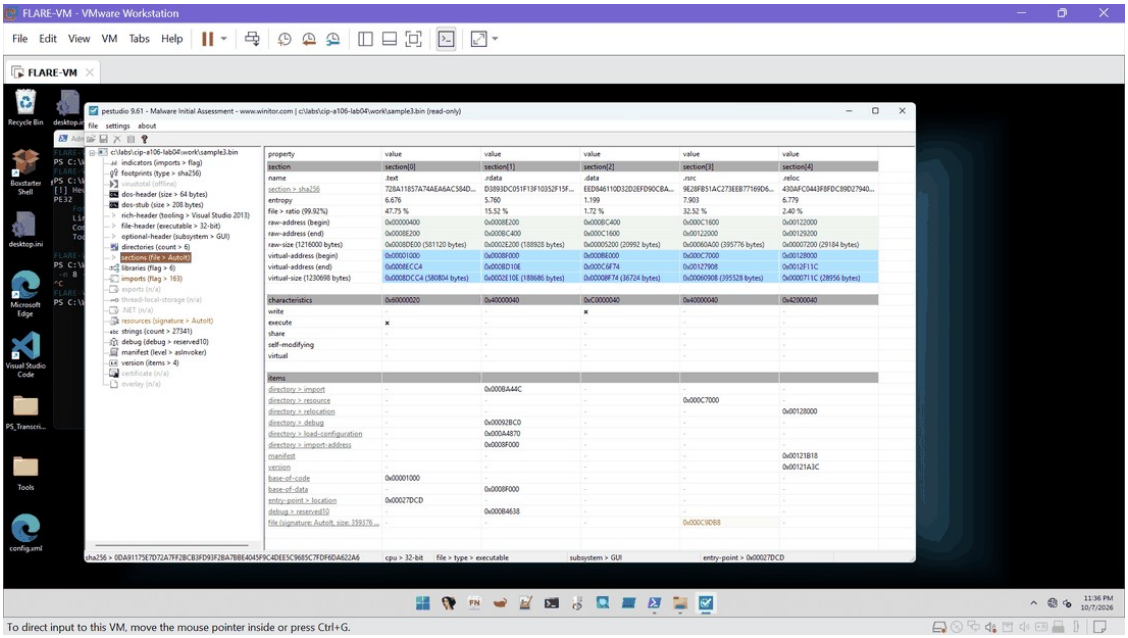

***Ex. E14*** | *2026-10-07 23:49:54 | VMware Workstation | Snapshot Manager: Lab04_PreDebug created 10/7/2026 11:49:54 PM (second snapshot, before any debugging) | SHA-256 EEAC3C035E498F16…*

## B. Static identification, PEStudio (Task A.4)

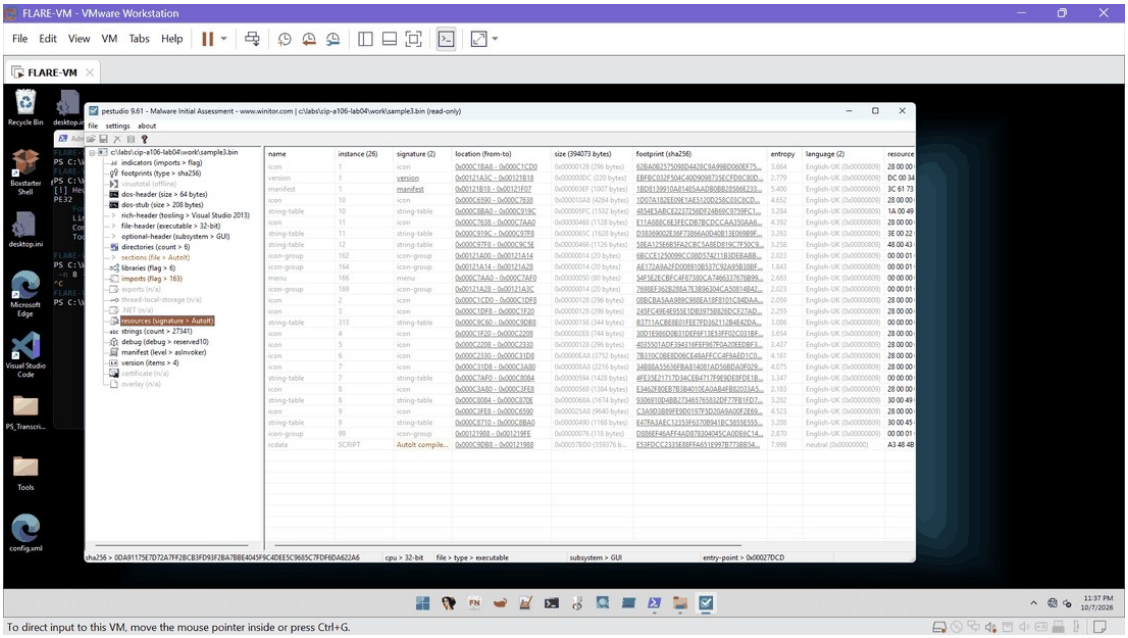
File header, sections and entropy, resources, imports, indicators and strings of the working copy.



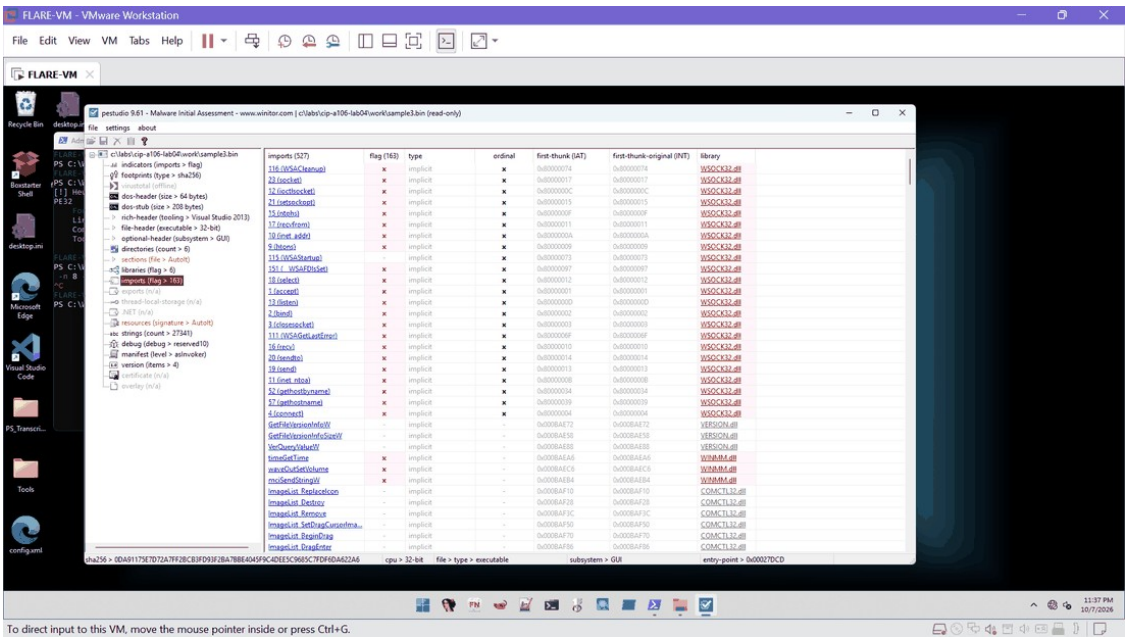

***Ex. E08*** | *2026-10-07 23:35 | PEStudio 9.61 | PEStudio header view of work\sample3.bin: SHA-256 0DA91175…622A6, size 1,217,024 bytes, entropy 7.199, 32-bit GUI executable, first 32 bytes MZ, entry point 0x27DCD (.text), signature string lists AutoIt compiled script, compiler stamp Wed Feb 26 11:25:12 2020 UTC, debug stamp Fri Jul 10 09:42:07 2015 UTC | SHA-256 A63427FAC1EE1CDF…*



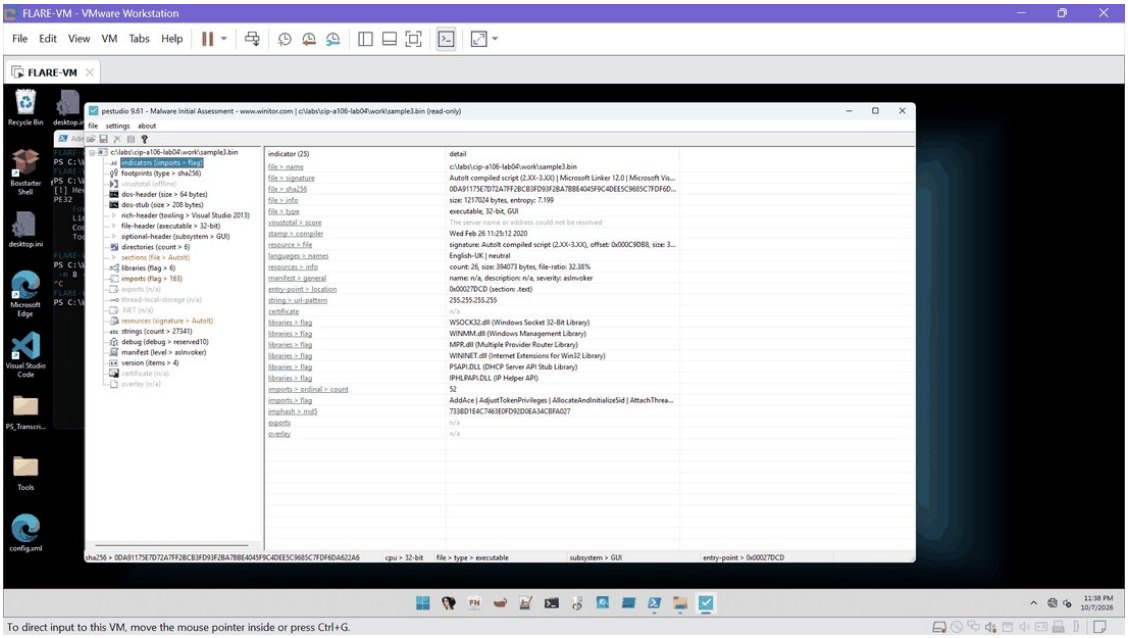

***Ex. E09*** | *2026-10-07 23:36 | PEStudio 9.61 | PEStudio sections: .text 6.676, .rdata 5.760, .data 1.199, .rsrc 7.903 (32.52 % of the file), .reloc 6.779; file signature AutoIt at 0xC9DB8 | SHA-256 258DC45996CDD428…*



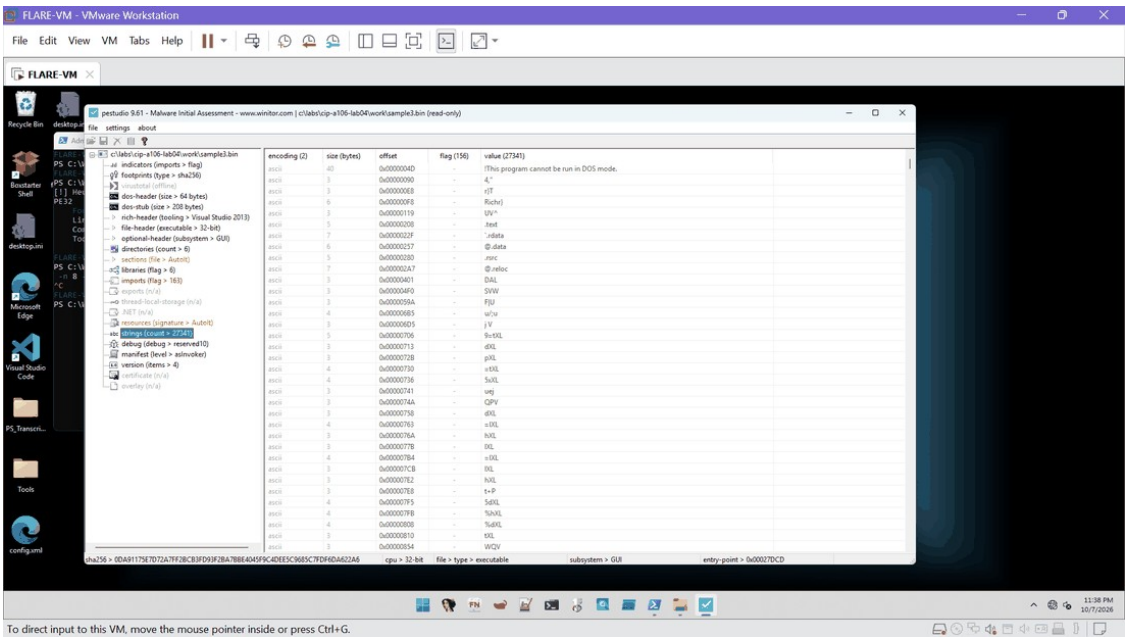

***Ex. E10*** | *2026-10-07 23:37 | PEStudio 9.61 | PEStudio resources: 26 resources; RCDATA SCRIPT of 359,376 bytes, entropy 7.999, signature "AutoIt compiled script" | SHA-256 CC96DC548F947F88…*




***Ex. E11*** | *2026-10-07 23:37* | *PEStudio 9.61* | *PEStudio imports (527, 163 flagged): WSOCK32 socket functions, VERSION, WINMM, COMCTL32 and others* | *SHA-256 3A6BB46F2A6D3FF0…*



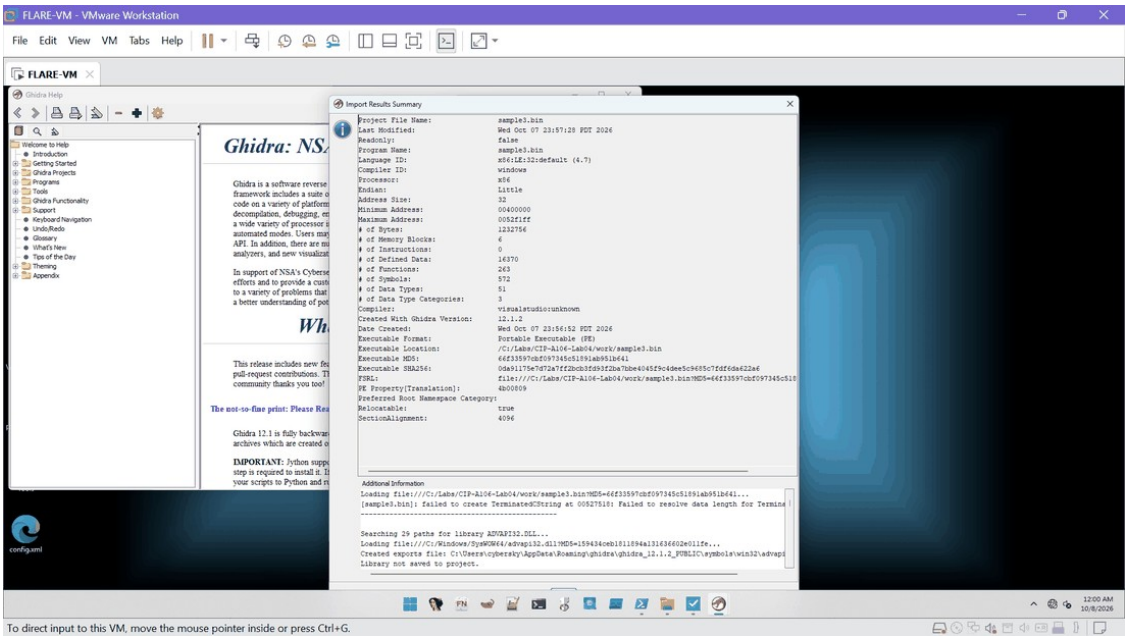

***Ex. E12*** | *2026-10-07 23:38* | *PEStudio 9.61* | *PEStudio indicators (25): AutoIt compiled script signature, libraries WSOCK32, WINMM, MPR, WININET, PSAPI, IPHLPAPI, string pattern 255.255.255.255, imphash MD5 733BD1E4C7463E0FD92D0EA34CBFA027* | *SHA-256 E19A19FCAFA1B40D…*



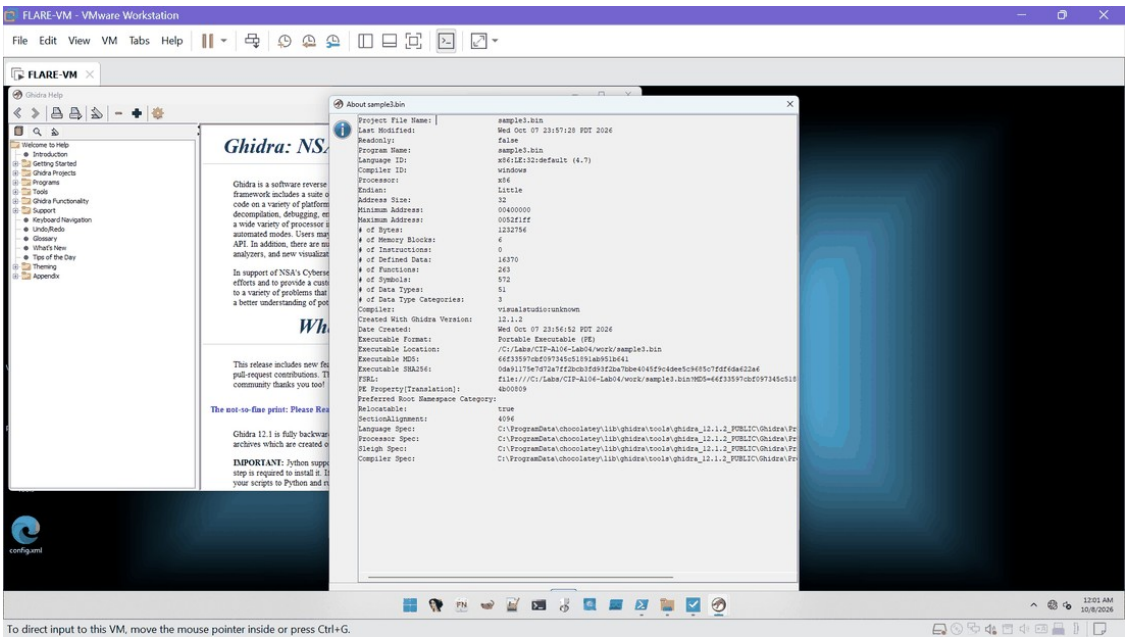

***Ex. E13*** | *2026-10-07 23:38* | *PEStudio 9.61* | *PEStudio strings view (27,341 strings): the first entries are compiler and section-name artefacts* | *SHA-256 6B4A1107425AD386…*

## C. Ghidra static analysis (Task C)

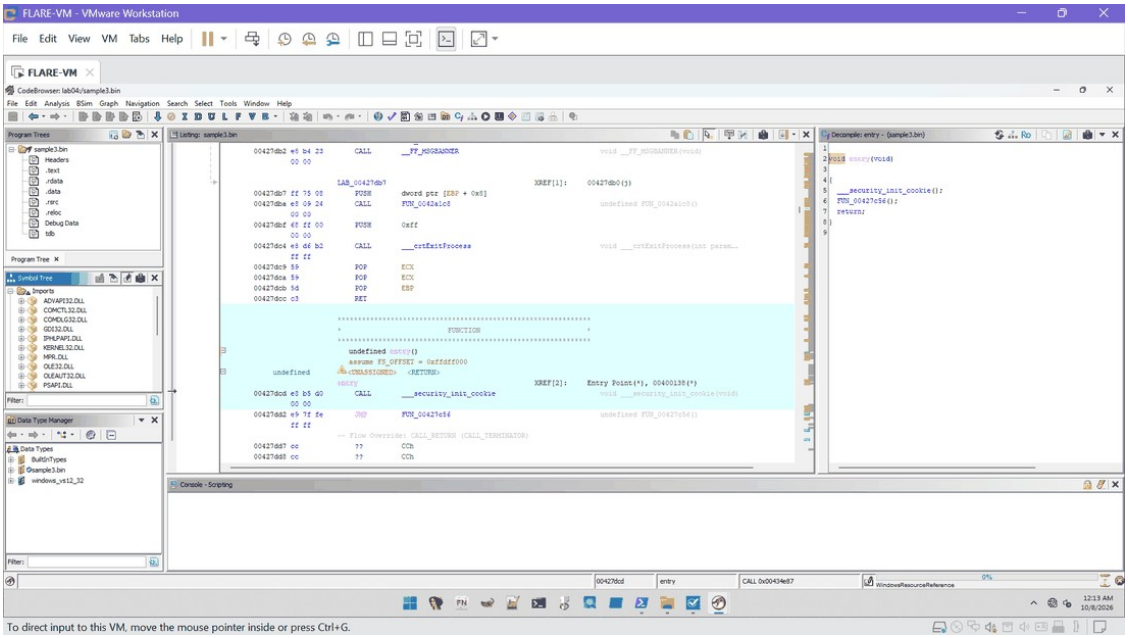
Project import and hash, program information, entry point, WININET imports, and the reference checks for the dialog string and IsDebuggerPresent.



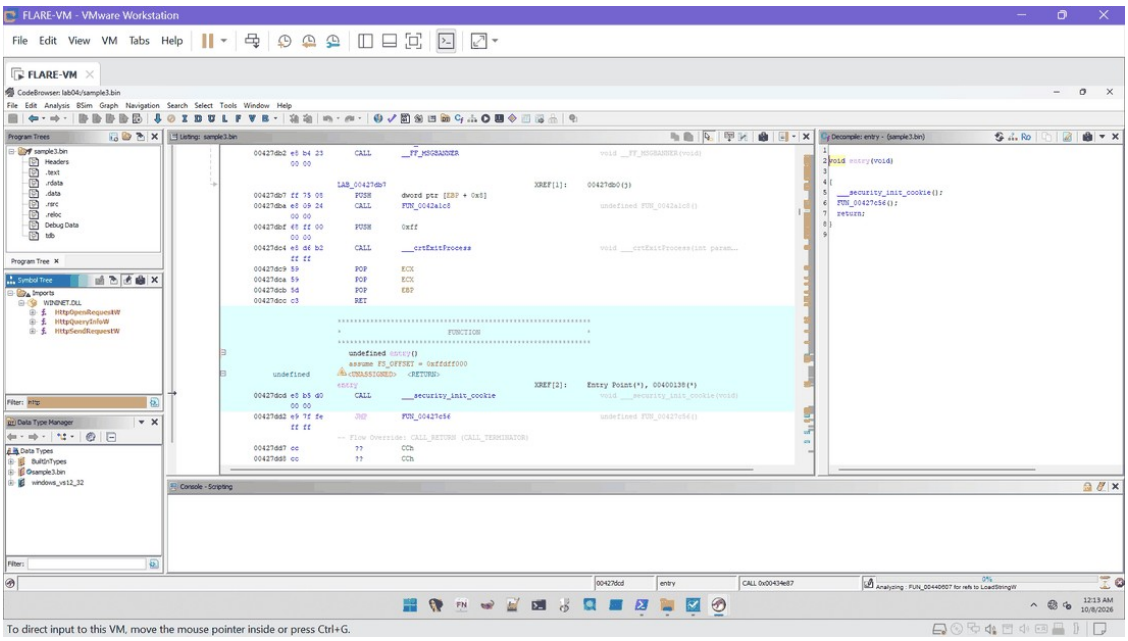

***Ex. E15*** | *2026-10-08 00:00* | *Ghidra 12.1.2* | *Ghidra Import Results Summary for sample3.bin: x86:LE:32:default, MD5 66f33597cbf097345c51891ab951b641, SHA256 0da91175…622a6, location C:/Labs/CIP-A106-Lab04/work/sample3.bin* | *SHA-256 FC4512A59117F6AF…*



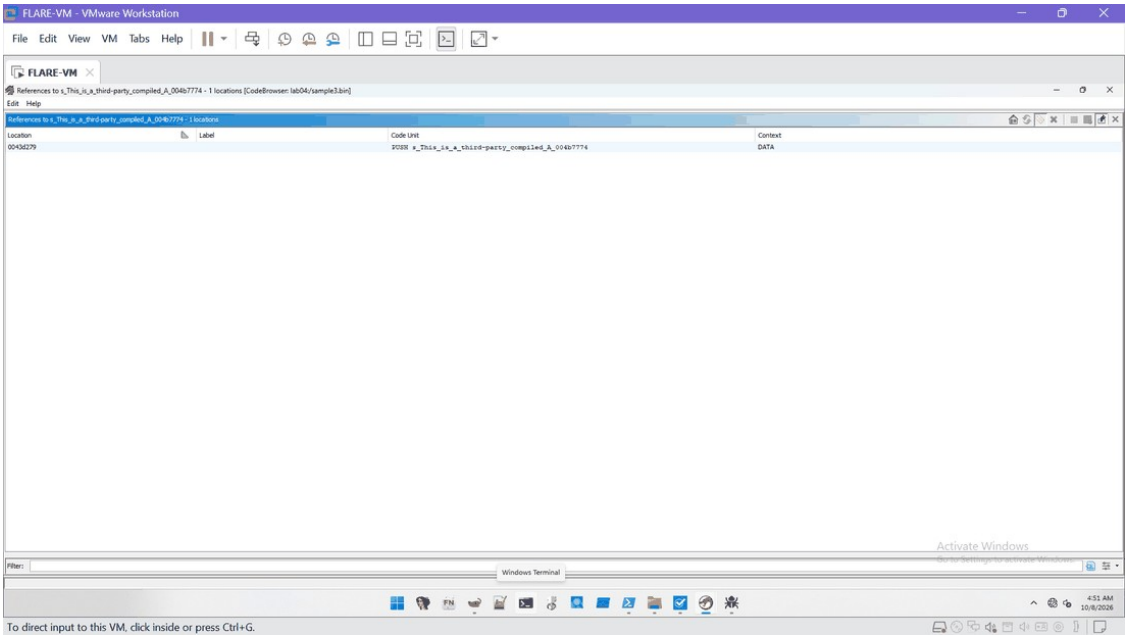

***Ex. E16*** | *2026-10-08 00:01 | Ghidra 12.1.2 | Ghidra program information after analysis: 263 functions, 572 symbols, 6 memory blocks, created with Ghidra 12.1.2, same MD5 and SHA256 | SHA-256 D8194FFE5B7A81F3…*



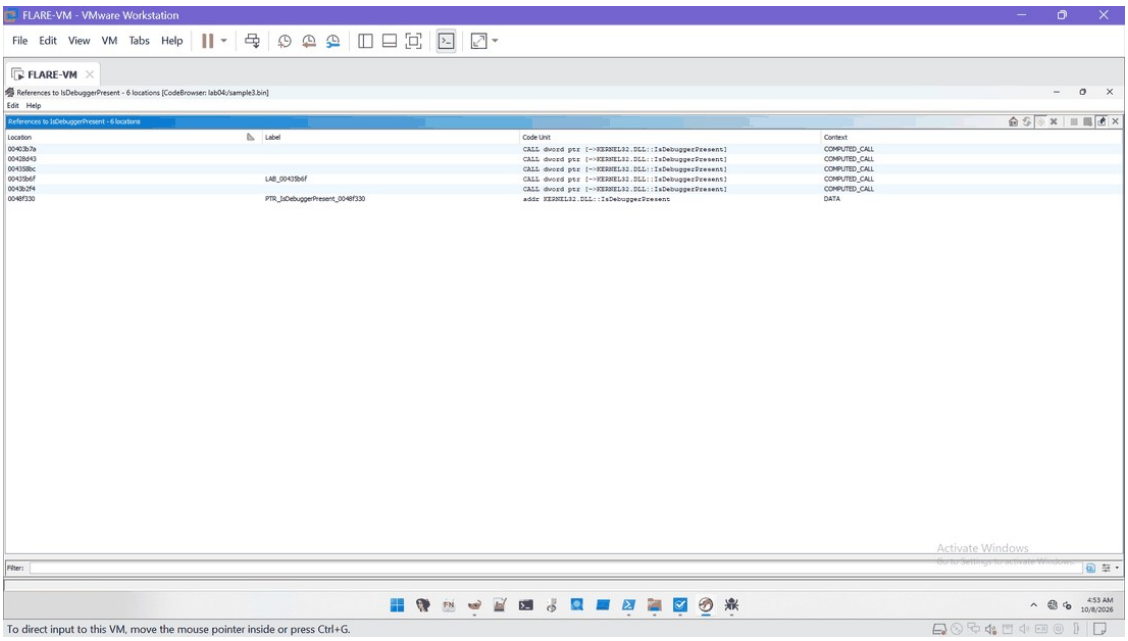

***Ex. E17*** | *2026-10-08 00:13 | Ghidra 12.1.2 | Ghidra CodeBrowser: entry at 00427dcd calls __security_init_cookie then jumps to FUN_00427c56 (standard MSVC startup) | SHA-256 5BD9C8C9842322AE…*



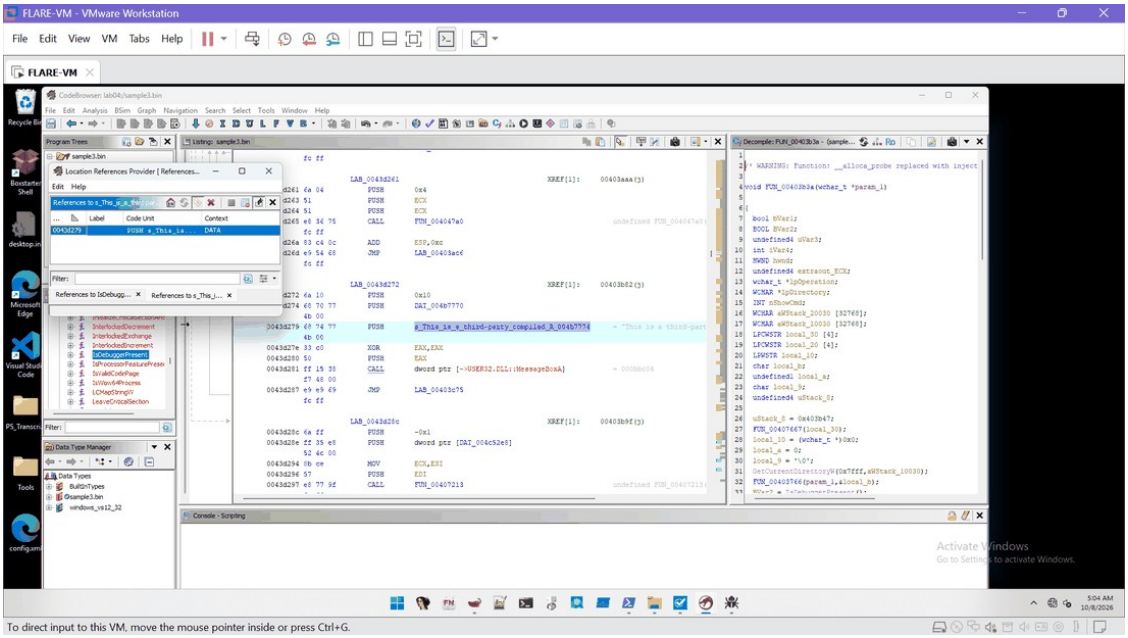

***Ex. E18*** | *2026-10-08 00:13 | Ghidra 12.1.2 | Ghidra Symbol Tree filtered on "http": WININET imports HttpOpenRequestW, HttpQueryInfoW, HttpSendRequestW | SHA-256 AC446C483204DAEB…*



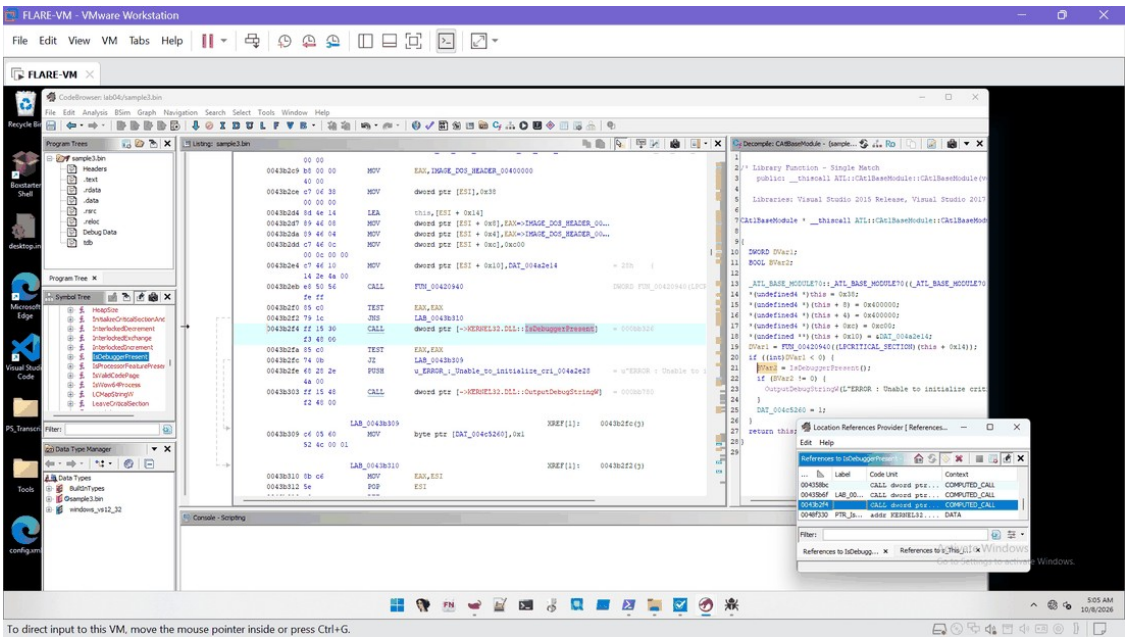

***Ex. E41*** | *2026-10-08 04:51 | Ghidra 12.1.2 | Ghidra references to the string "This is a third-party compiled AutoIt script.": one location, 0043d279 (PUSH) | SHA-256 E9196209DC13BB6F…*



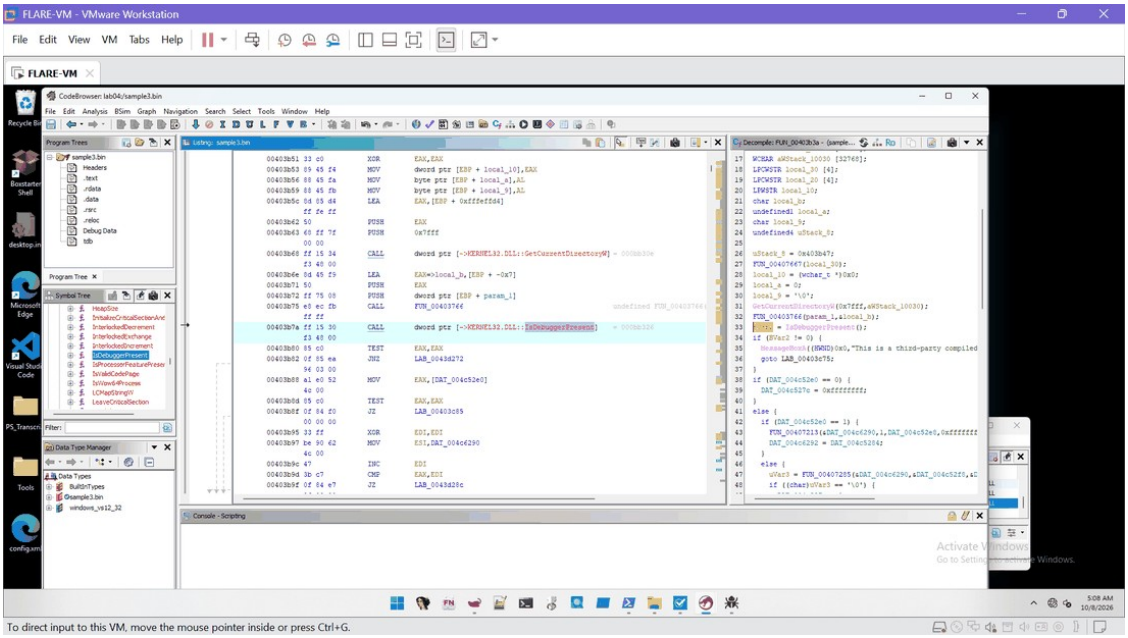

***Ex. E42*** | *2026-10-08 04:53* | *Ghidra 12.1.2* | *Ghidra references to IsDebuggerPresent: six locations (00403b7a, 00428d43, 004358bc, 00435b6f, 0043b2f4 and the import pointer 0048f330)* | *SHA-256 F775B6A8F409E24F…*



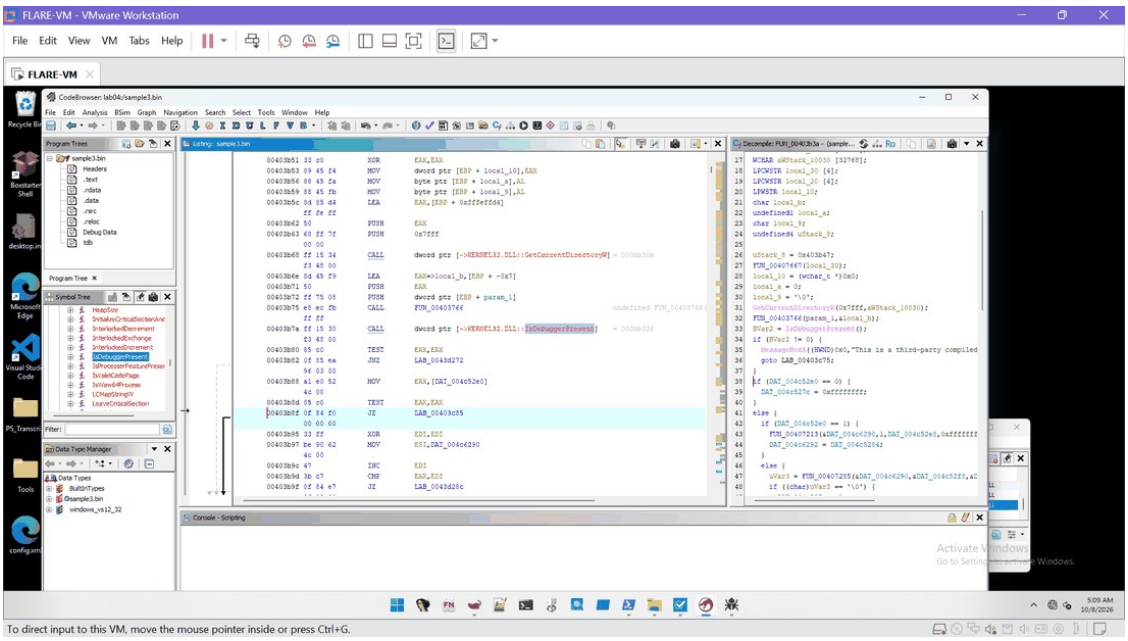

***Ex. E43*** | *2026-10-08 05:04* | *Ghidra 12.1.2* | *Ghidra listing at 0043d272 to 0043d281: PUSH of the dialog string then CALL MessageBoxA; decompile of FUN_00403b3a on the right* | *SHA-256 8C4166B20759976B…*




***Ex. E44*** | *2026-10-08 05:05 | Ghidra 12.1.2 | Ghidra call site 0043b2f4 inside CAtlBaseModule (library function, single match, Visual Studio 2015/2017): a benign ATL initialisation check, not a sample-specific anti-debug check | SHA-256 8520CFB8D91AB524…*



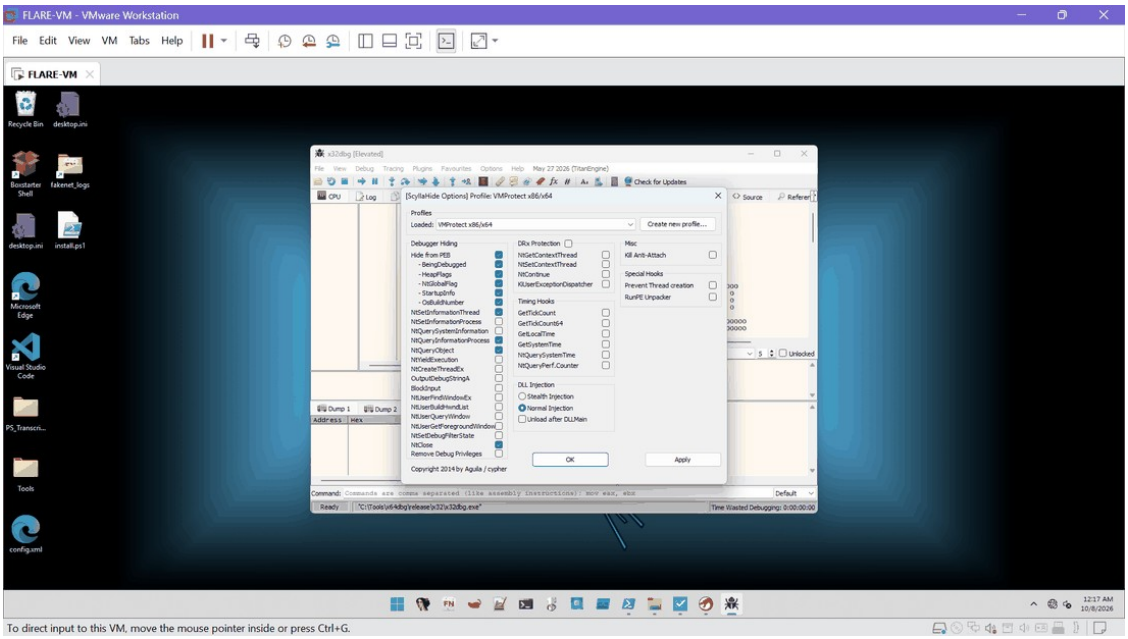

***Ex. E45*** | *2026-10-08 05:08 | Ghidra 12.1.2 | Ghidra FUN_00403b3a: CALL IsDebuggerPresent at 00403b7a; decompile shows BVar2 = IsDebuggerPresent(); if (BVar2 != 0) MessageBoxA(…); goto exit | SHA-256 F179DDB0DBAE4390…*



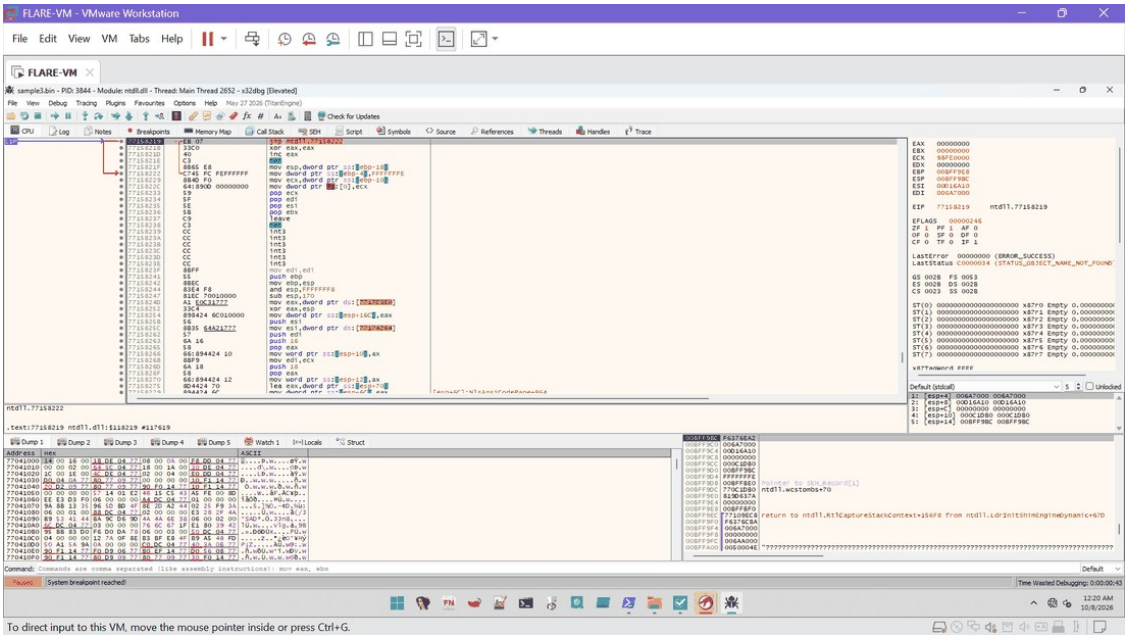

***Ex. E46*** | *2026-10-08 05:09 | Ghidra 12.1.2 | Ghidra FUN_00403b3a: TEST EAX,EAX at 00403b80, JNZ to LAB_0043d272 at 00403b82 (the branch taken when a debugger is reported) | SHA-256 1A65085FADD2DACD…*

# D. Debugger observations (Task D)

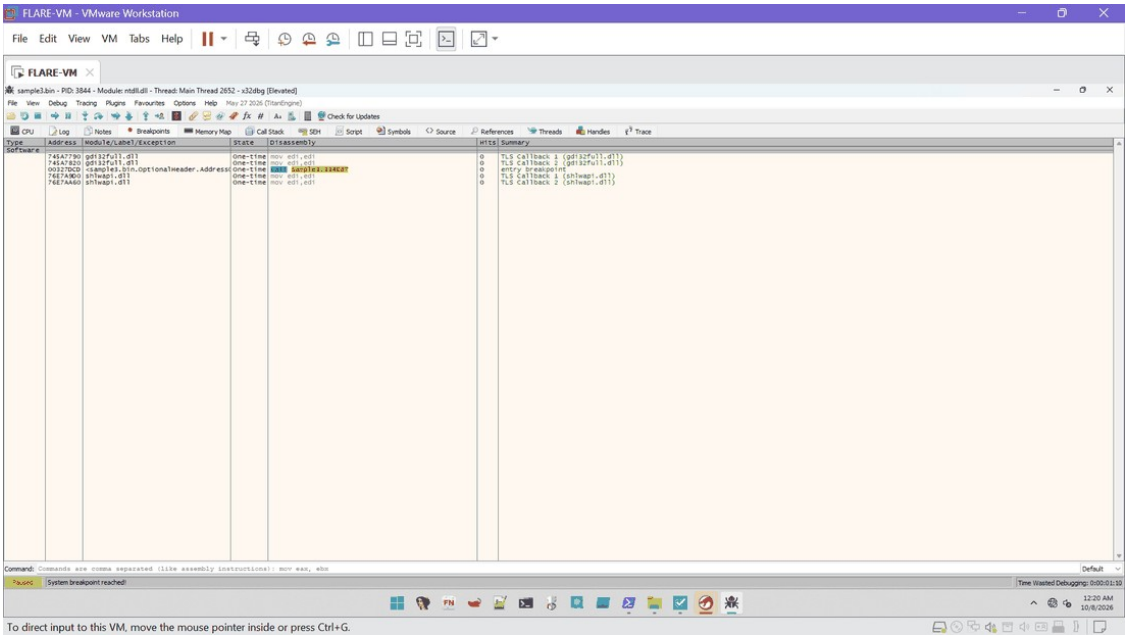
x32dbg runs 1, 2 and 3: ScyllaHide settings, breakpoints, system and entry stops, IsDebuggerPresent hit, the AutoIt dialog, call stack and logs.



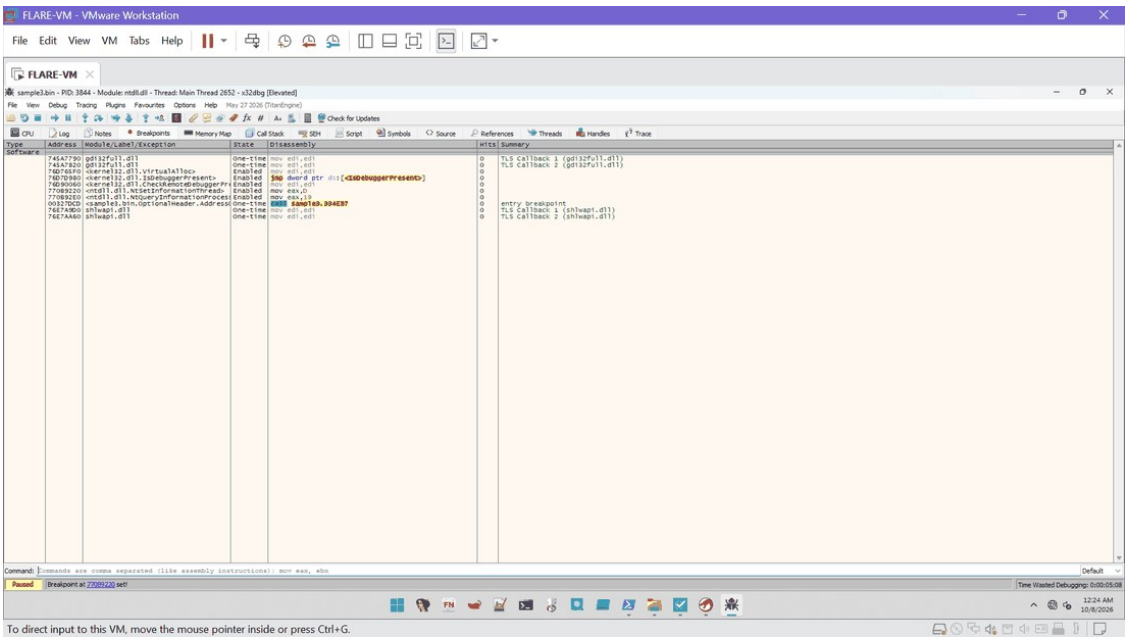

***Ex. E19*** | *2026-10-08 00:17 | x32dbg / ScyllaHide | ScyllaHide Options as found (profile VMProtect x86/x64): Hide from PEB items, NtSetInformationThread, NtQueryInformationProcess, NtQueryObject and NtClose ticked | SHA-256 426649D9A90FC3B9…*



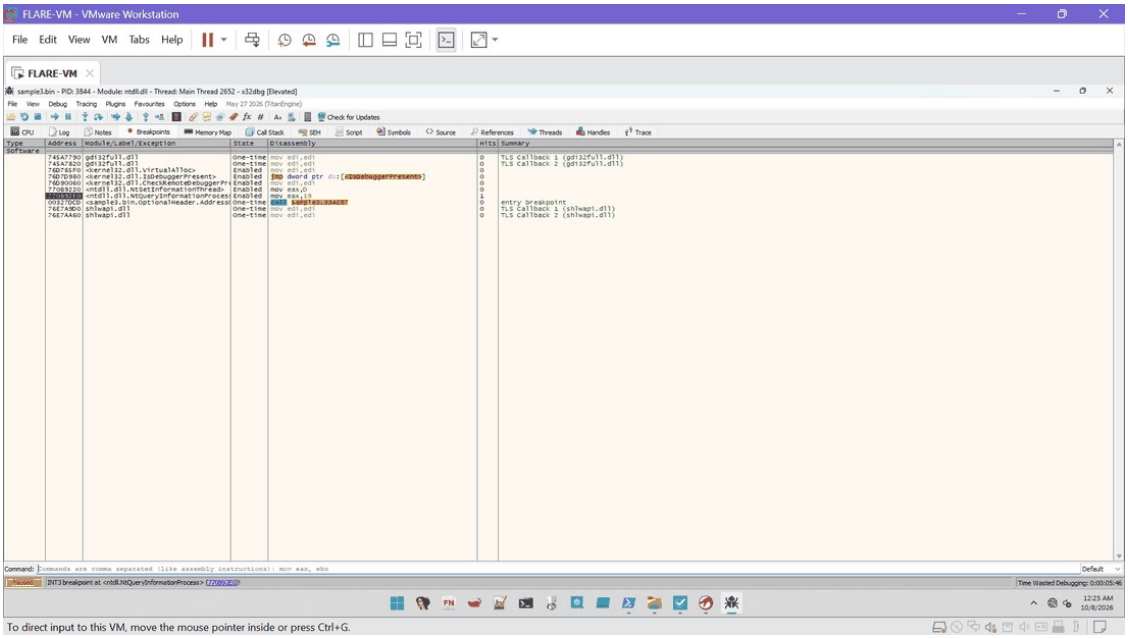

***Ex. E20*** | *2026-10-08 00:20* | *x32dbg* | *Run 1, PID 3844: system breakpoint reached in ntdll (EIP 77158219); first stop after process creation* | *SHA-256 31D8B8BE26AF9A3F…*



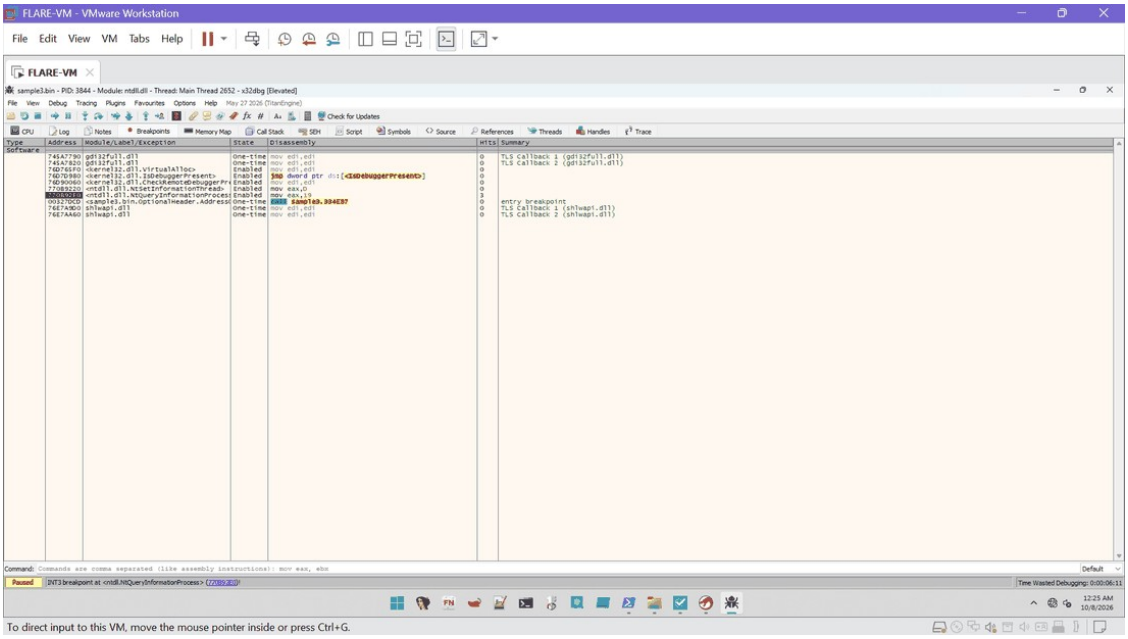

***Ex. E21*** | *2026-10-08 00:20* | *x32dbg* | *Run 1 Breakpoints tab at the system breakpoint: only TLS callback and entry breakpoints, all hit counts 0* | *SHA-256 10E47C6D63748BA3…*



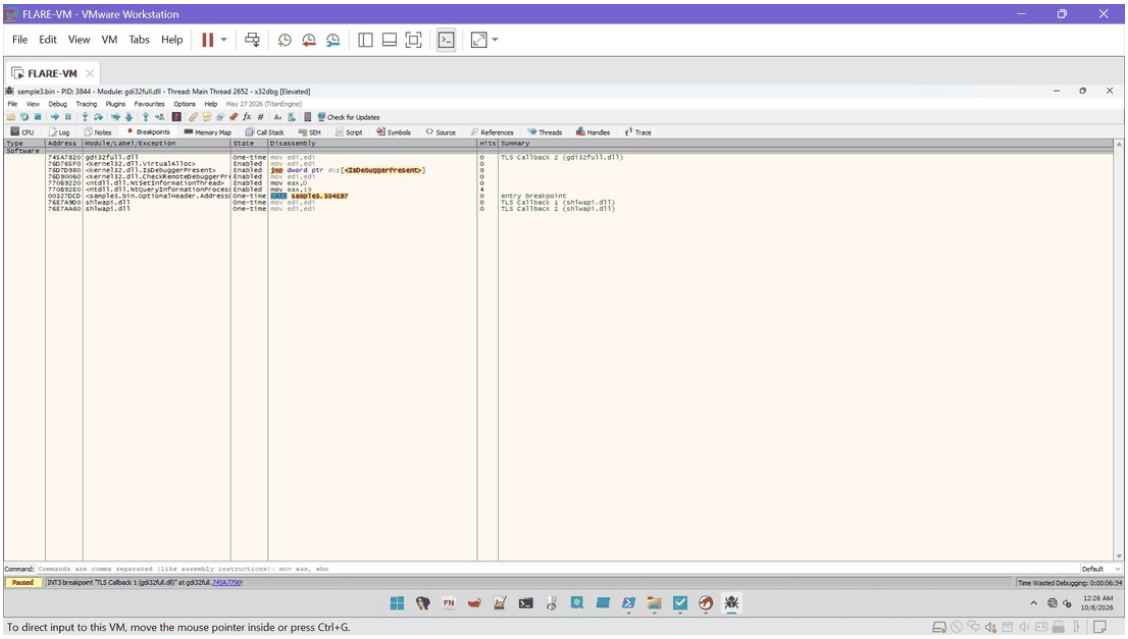

***Ex. E22*** | *2026-10-08 00:24 | x32dbg | Run 1 Breakpoints tab: VirtualAlloc 76D765F0, IsDebuggerPresent 76D7D980, CheckRemoteDebuggerPresent 76D90060, NtSetInformationThread 770B9220, NtQueryInformationProcess 770B92E0 and entry breakpoint 00327DCD (observation breakpoints, no code changed) | SHA-256 1991341E418945B2…*



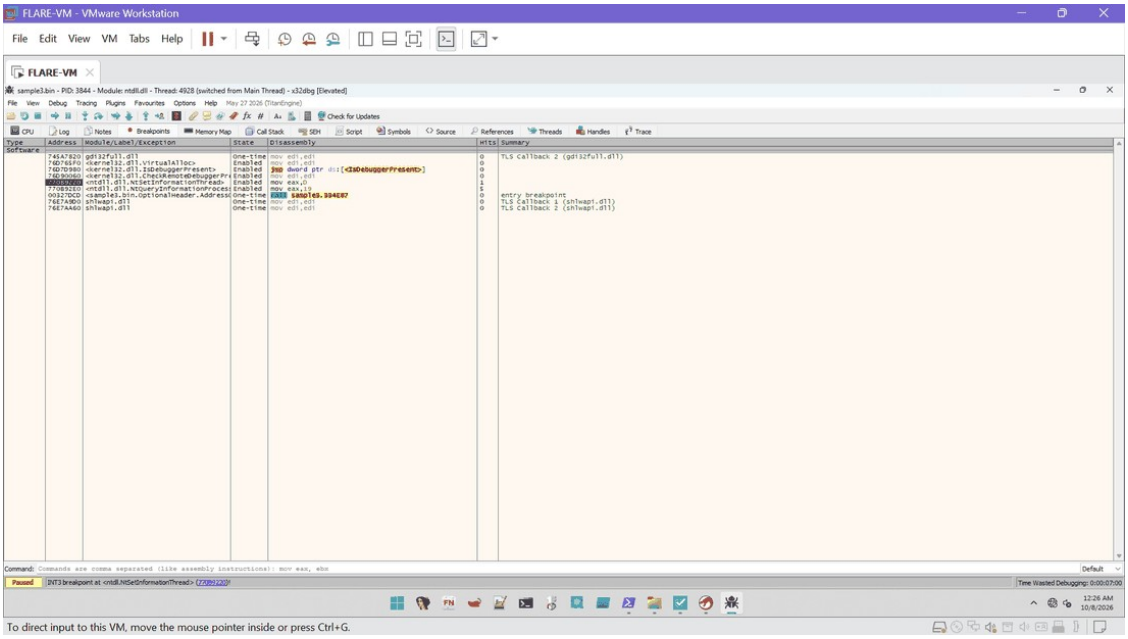

***Ex. E23*** | *2026-10-08 00:25 | x32dbg | Run 1: INT3 breakpoint at ntdll.NtQueryInformationProcess (770B92E0), hit count 1 | SHA-256 F004CED93B6C7987…*



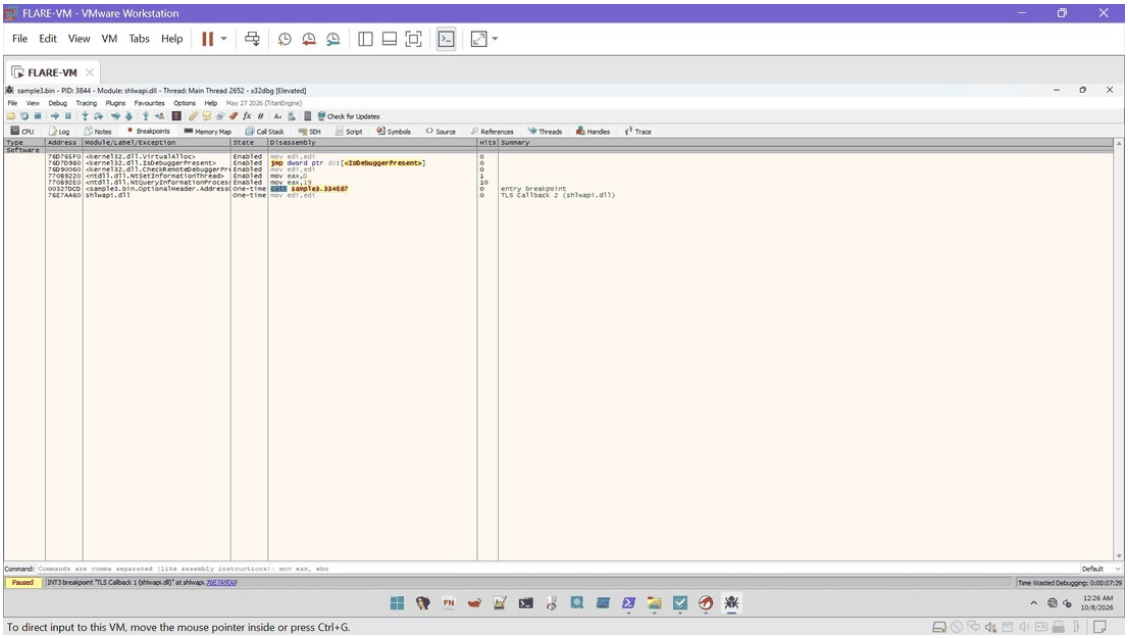

***Ex. E24*** | *2026-10-08 00:25* | *x32dbg* | *Run 1: NtQueryInformationProcess hit count 3* | *SHA-256 EA2E0F6E8251CA89…*



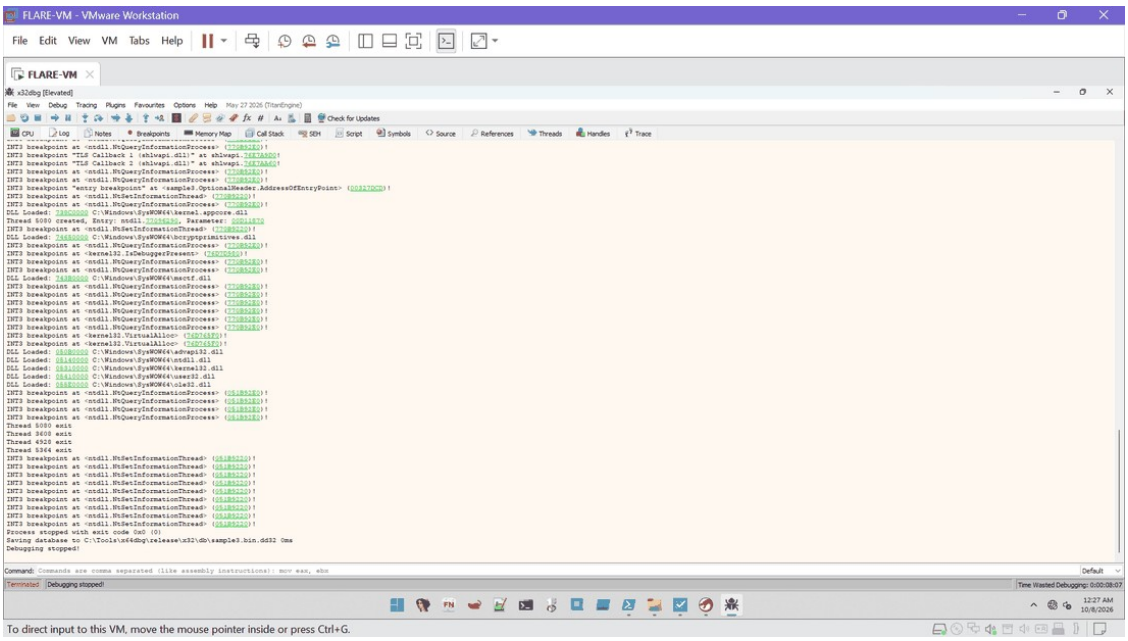

***Ex. E25*** | *2026-10-08 00:26* | *x32dbg* | *Run 1: TLS callback 1 (gdi32full) stop; NtQueryInformationProcess hit count 4* | *SHA-256 37FC54D3E063785A…*



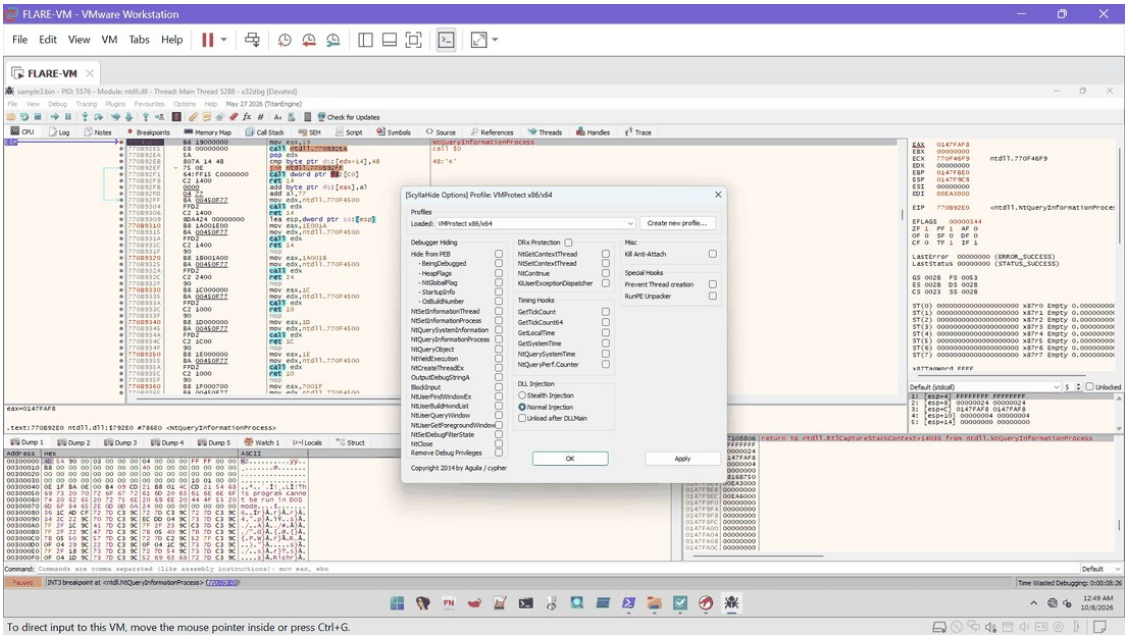

***Ex. E26*** | *2026-10-08 00:26 | x32dbg | Run 1: breakpoint at ntdll.NtSetInformationThread (770B9220) from thread 4928, hit count 1 | SHA-256 8F19CF007BA1CC11…*



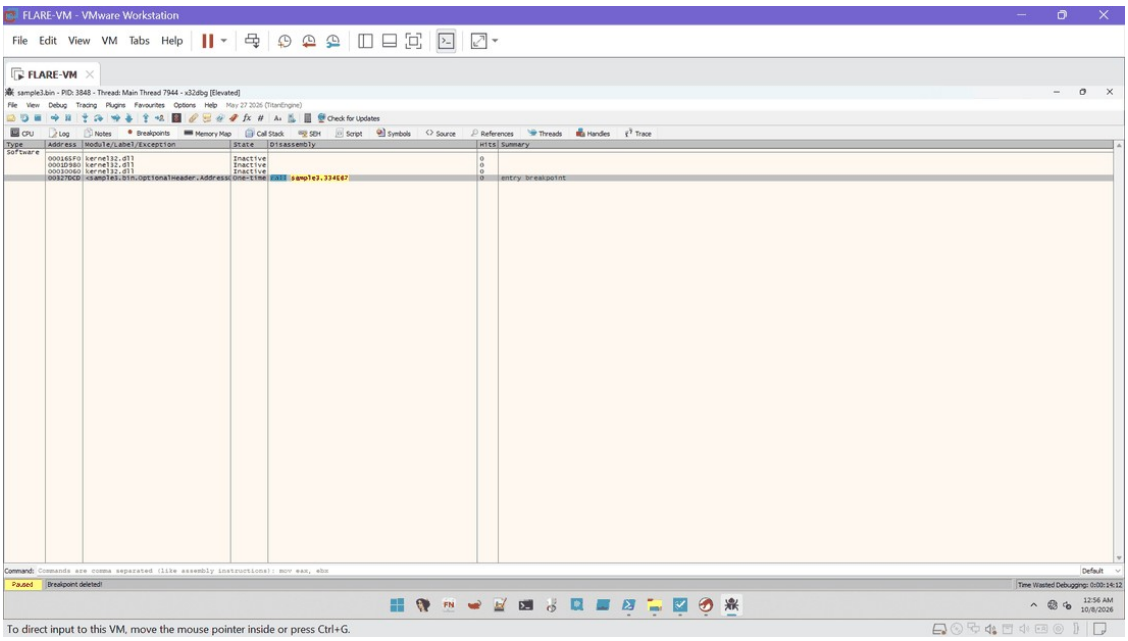

***Ex. E27*** | *2026-10-08 00:26 | x32dbg | Run 1: TLS callback 1 (shlwapi) stop; NtQueryInformationProcess hit count 10 | SHA-256 C8898D8E46912630…*



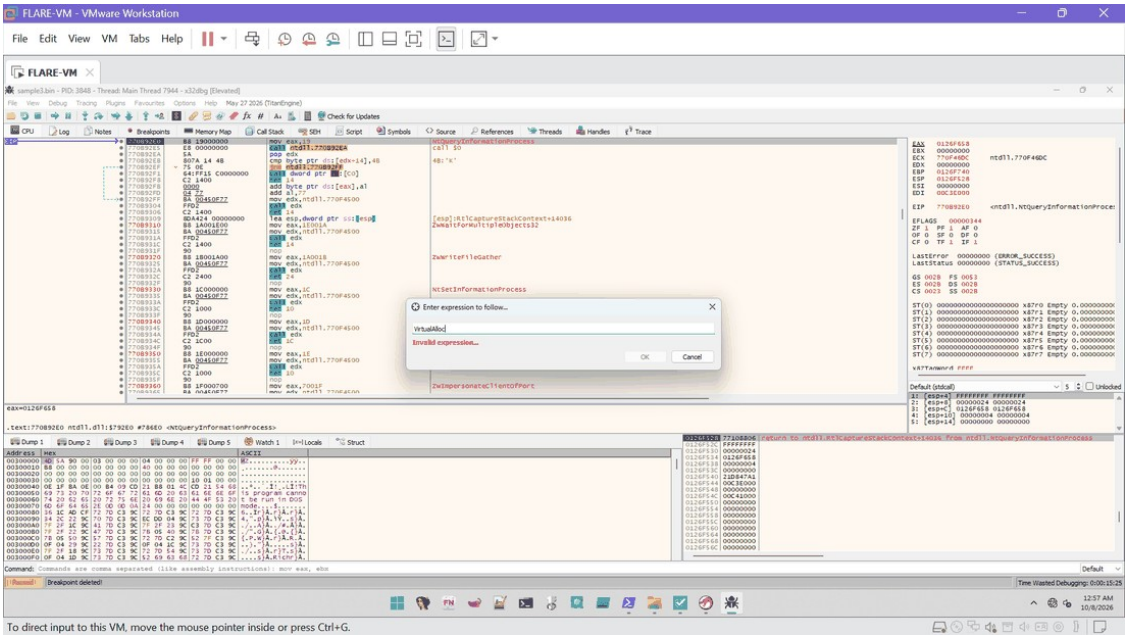

***Ex. E28*** | *2026-10-08 00:27 | x32dbg | Run 1 final log: two VirtualAlloc breakpoint lines (76D765F0), five private copies of advapi32, ntdll, kernel32, user32 and ole32 loaded from SysWOW64, NtQueryInformationProcess at 051B92E0, NtSetInformationThread at 051B9220, process stopped with exit code 0x0 | SHA-256 49B6DC3CC58B2AF9…*



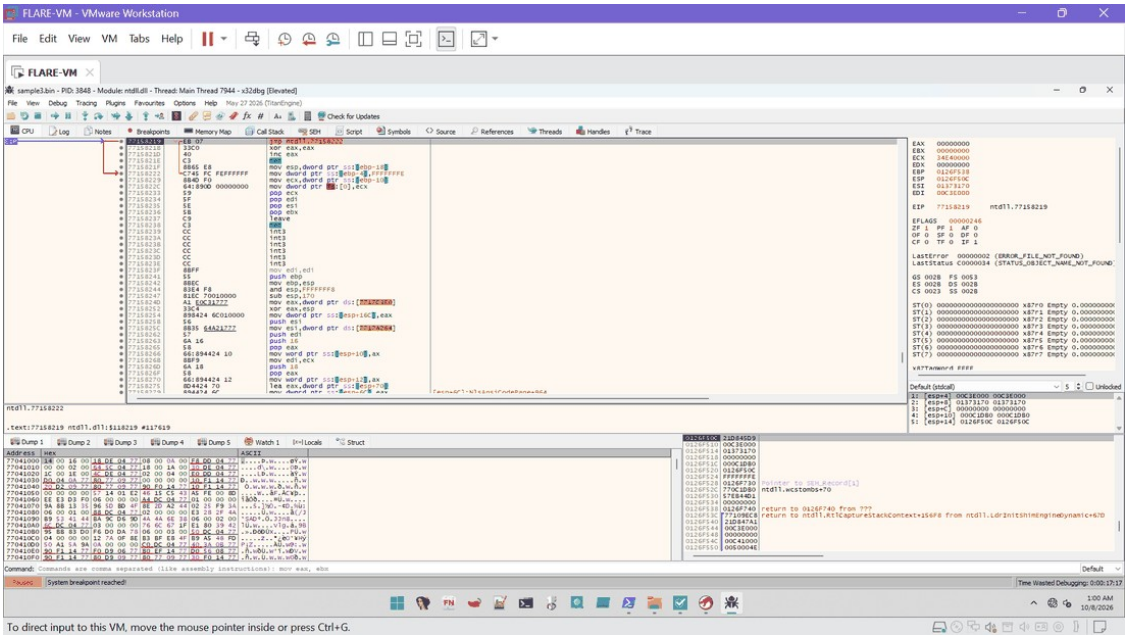

***Ex. E29*** | *2026-10-08 00:49 | x32dbg / ScyllaHide | Aborted start, PID 5576: ScyllaHide options changed after launch (all unticked); this start was killed and is not used as evidence | SHA-256 BA234FEDDF9D9B31…*



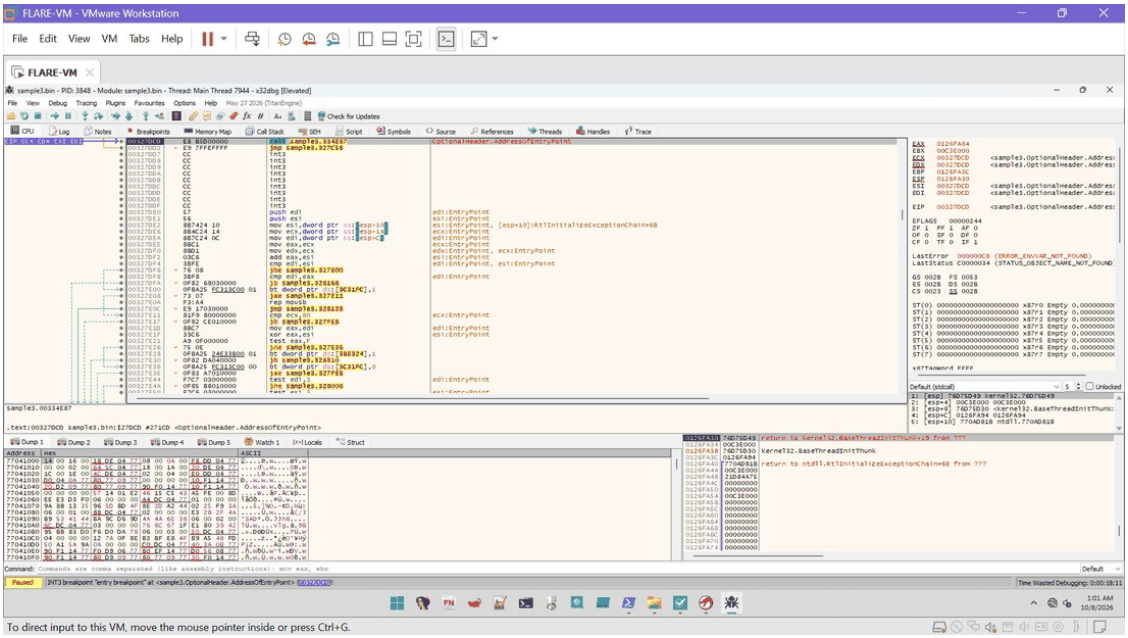

***Ex. E30*** | *2026-10-08 00:56 | x32dbg | Run 2, PID 3848: Breakpoints tab, kernel32 breakpoints Inactive until the module loads; entry breakpoint 00327DCD | SHA-256 3A0217C83B233EE5…*



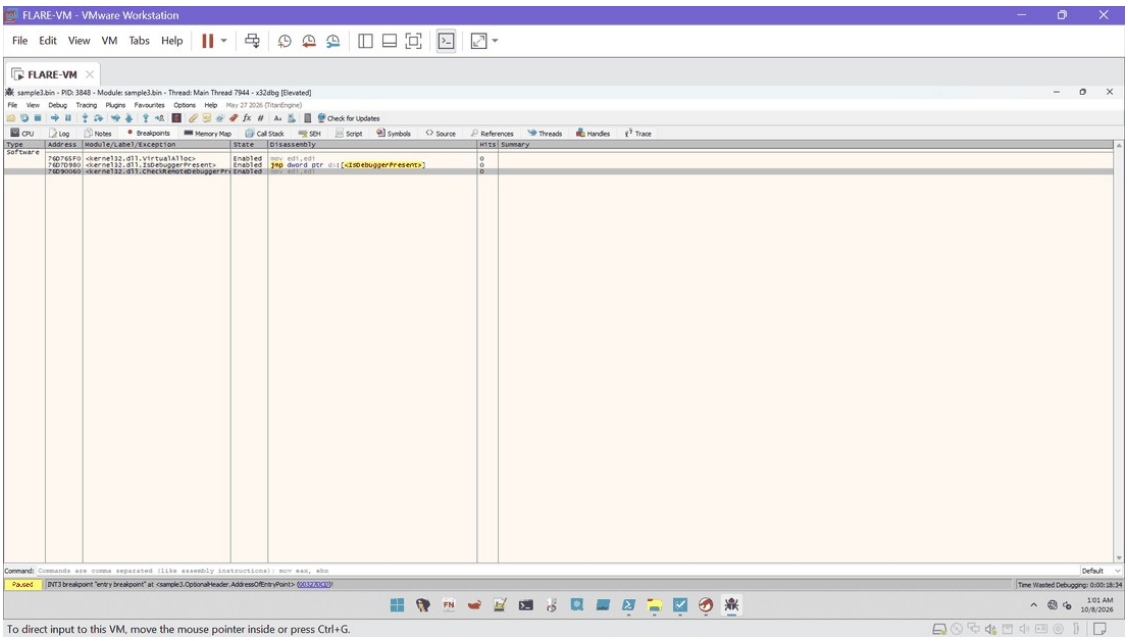

***Ex. E31*** | *2026-10-08 00:57 | x32dbg | Run 2: "Enter expression to follow" with VirtualAlloc returns Invalid expression (kernel32 not yet loaded at the system stop) | SHA-256 901F4C9B30528A8D…*



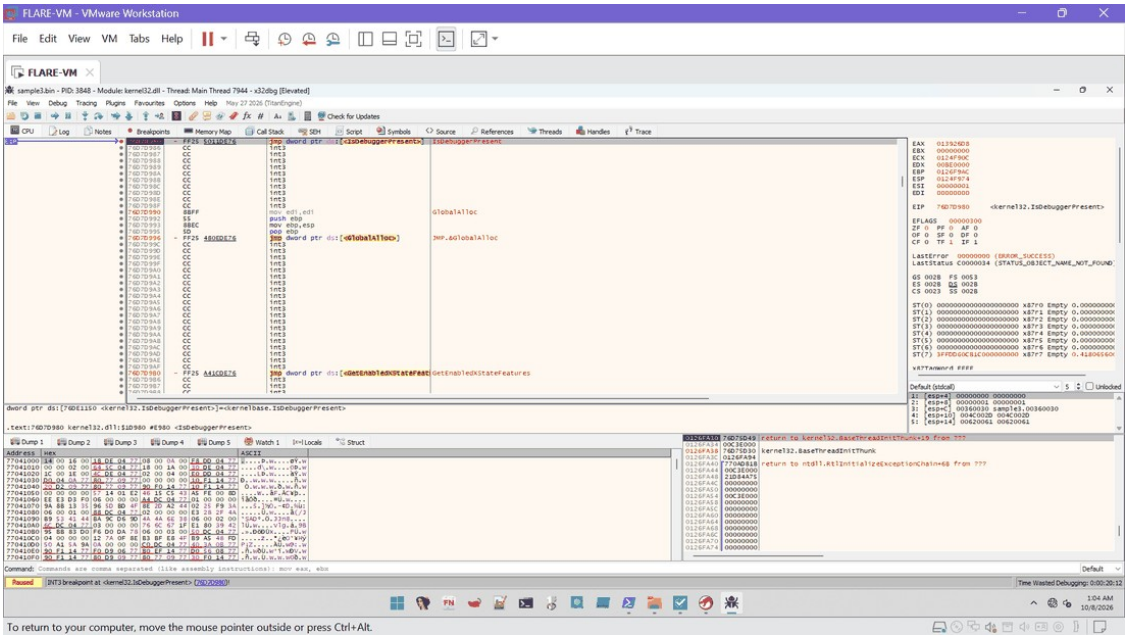

***Ex. E32*** | *2026-10-08 01:00 | x32dbg | Run 2: system breakpoint reached (PID 3848, ntdll 77158219) | SHA-256 A4DFC85CF6DB0AAC…*



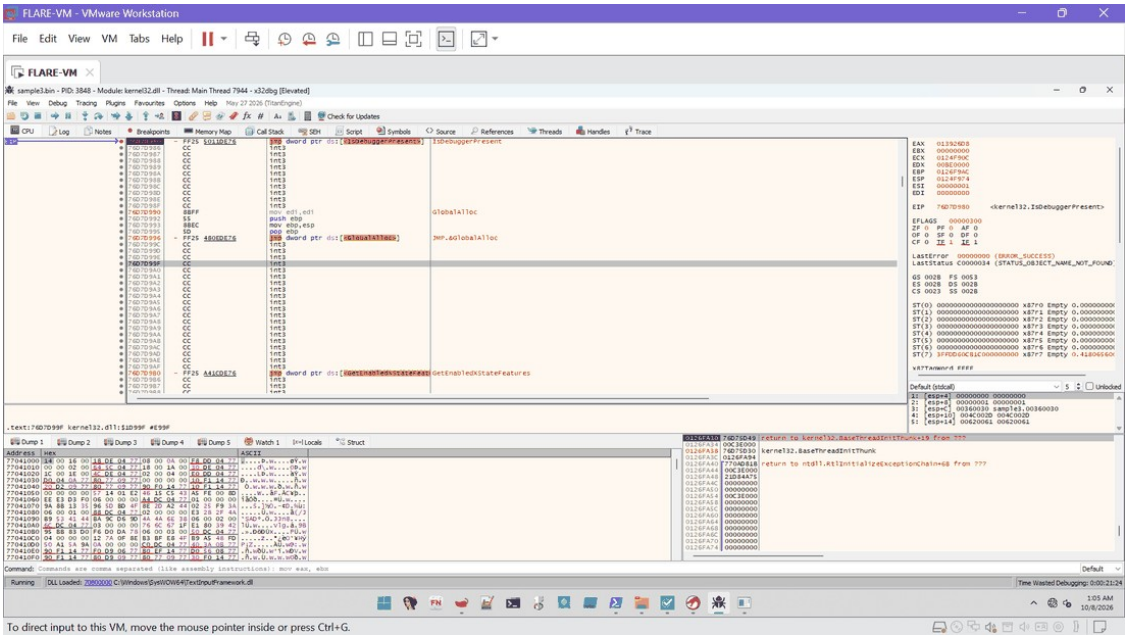

***Ex. E33*** | *2026-10-08 01:01 | x32dbg | Run 2: entry breakpoint at sample3.00327DCD; CALL sample3.334E87 at the entry point | SHA-256 C4C74F5A88CD9546…*



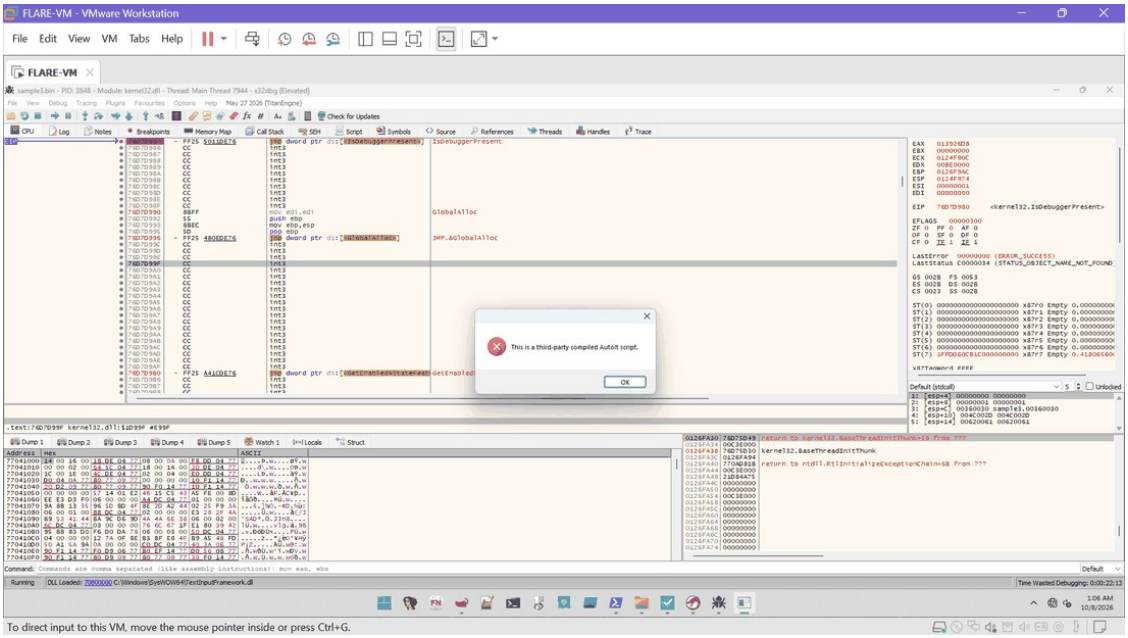

***Ex. E34*** | *2026-10-08 01:01 | x32dbg | Run 2 Breakpoints tab: kernel32.VirtualAlloc, IsDebuggerPresent and CheckRemoteDebuggerPresent Enabled with 0 hits | SHA-256 C22E4947051A2EFC…*



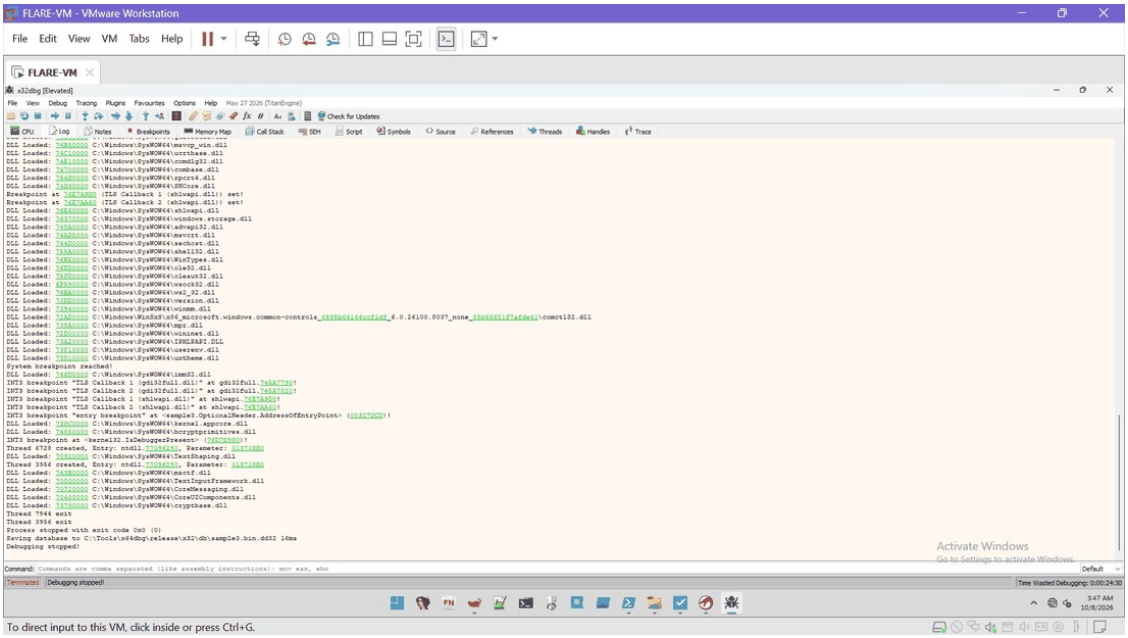

***Ex. E35*** | *2026-10-08 01:04 | x32dbg | Run 2: INT3 breakpoint at kernel32.IsDebuggerPresent (76D7D980), first instruction JMP dword ptr [IsDebuggerPresent], called from the sample | SHA-256 1769CB8E622B32FA…*



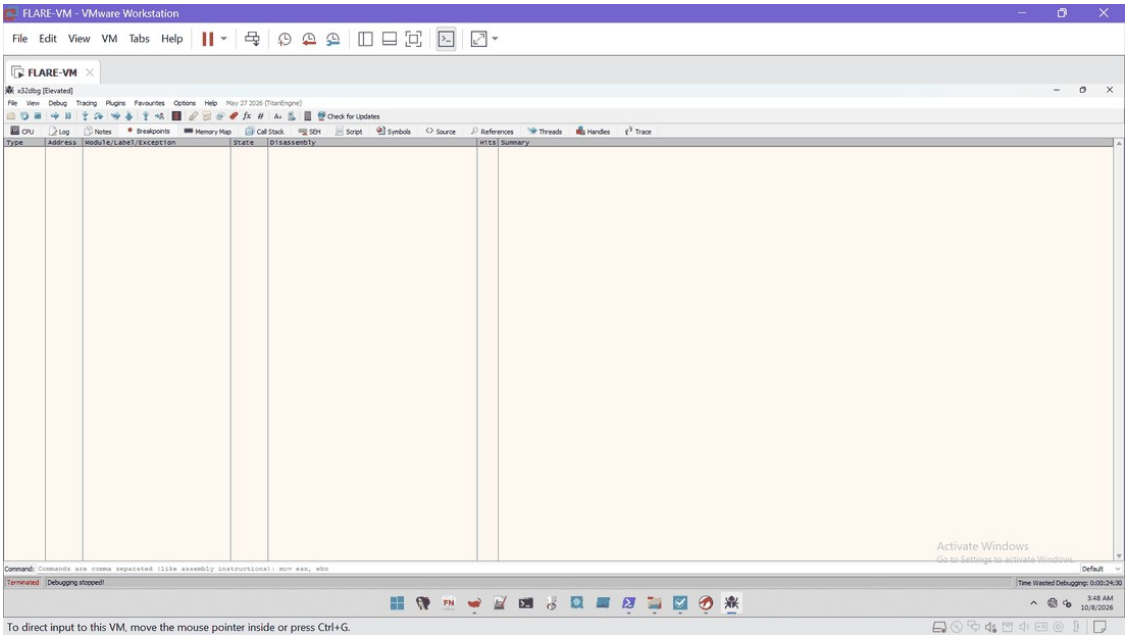

***Ex. E36*** *| 2026-10-08 01:05 | x32dbg | Run 2: process continued (Running), TextInputFramework.dll loaded | SHA-256 B52960D6F250897F…*



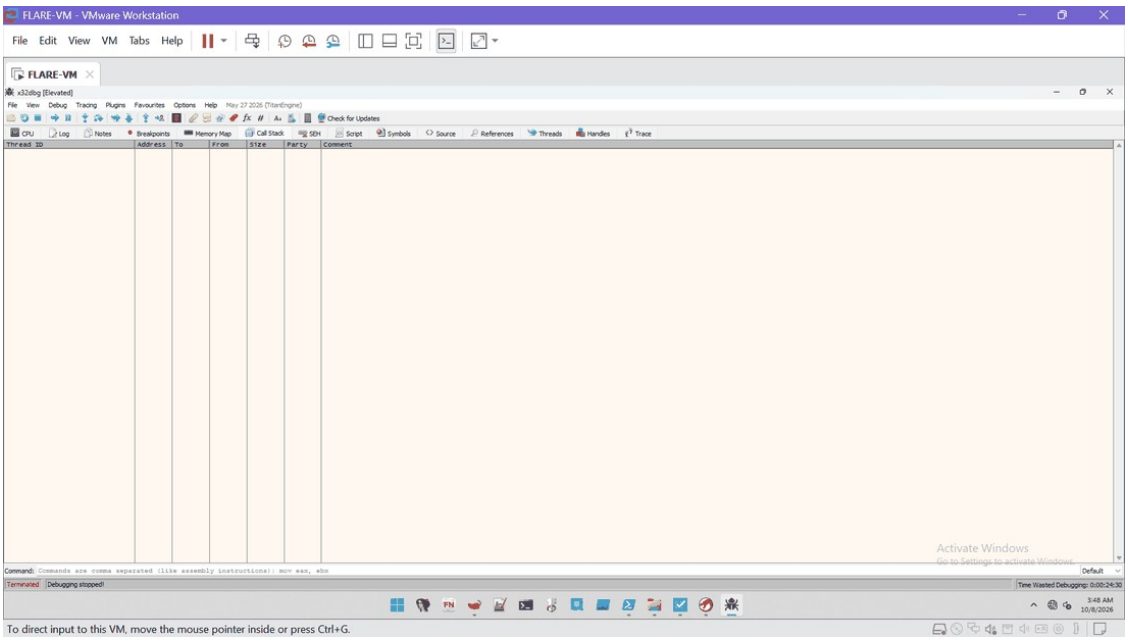

***Ex. E37*** *| 2026-10-08 01:06 | x32dbg | Run 2: message box "This is a third-party compiled AutoIt script." shown while the debugger is attached | SHA-256 CC829E716B545417…*



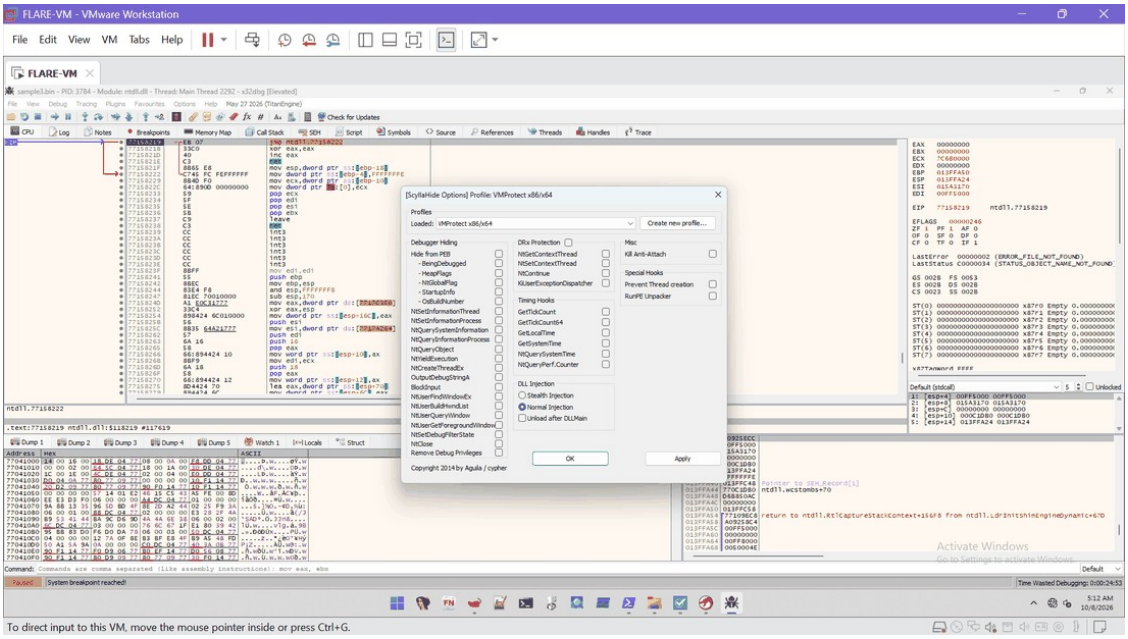

***Ex. E38*** | *2026-10-08 03:47 | x32dbg | Run 2 final log: IsDebuggerPresent breakpoint at 76D7D980, threads, process stopped with exit code 0x0, database saved; no VirtualAlloc line | SHA-256 0DD5E7DC7C96FEA8…*



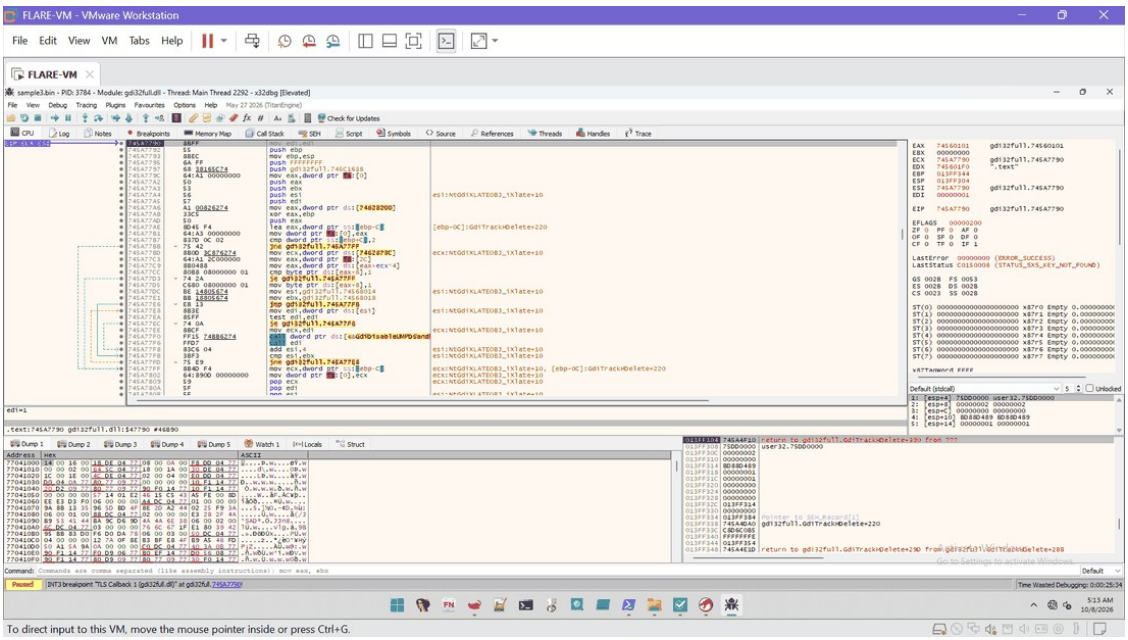

***Ex. E39*** | *2026-10-08 03:48 | x32dbg | After run 2: Breakpoints tab empty (session terminated) | SHA-256 9061C14FF999391C…*



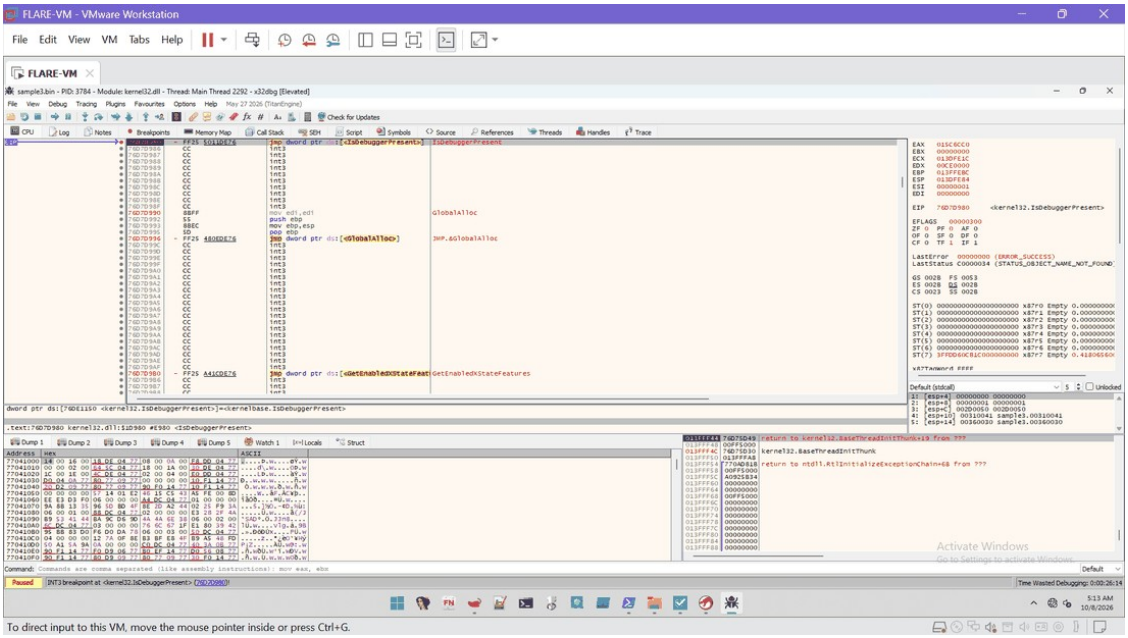

***Ex. E40*** | *2026-10-08 03:48* | *x32dbg* | *After run 2: Threads tab empty (session terminated)* | *SHA-256 17A50E45BE9C0AF4…*



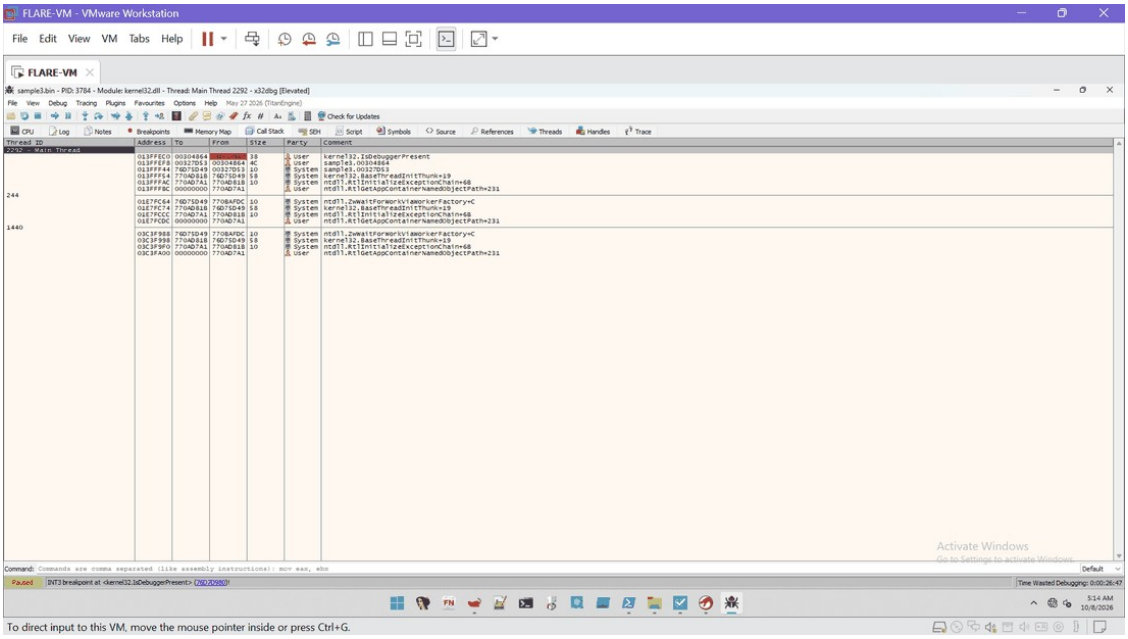

***Ex. E47*** | *2026-10-08 05:12* | *x32dbg / ScyllaHide* | *Run 3 at the system breakpoint: ScyllaHide options with all boxes unticked* | *SHA-256 7529C6E1BA43C700…*



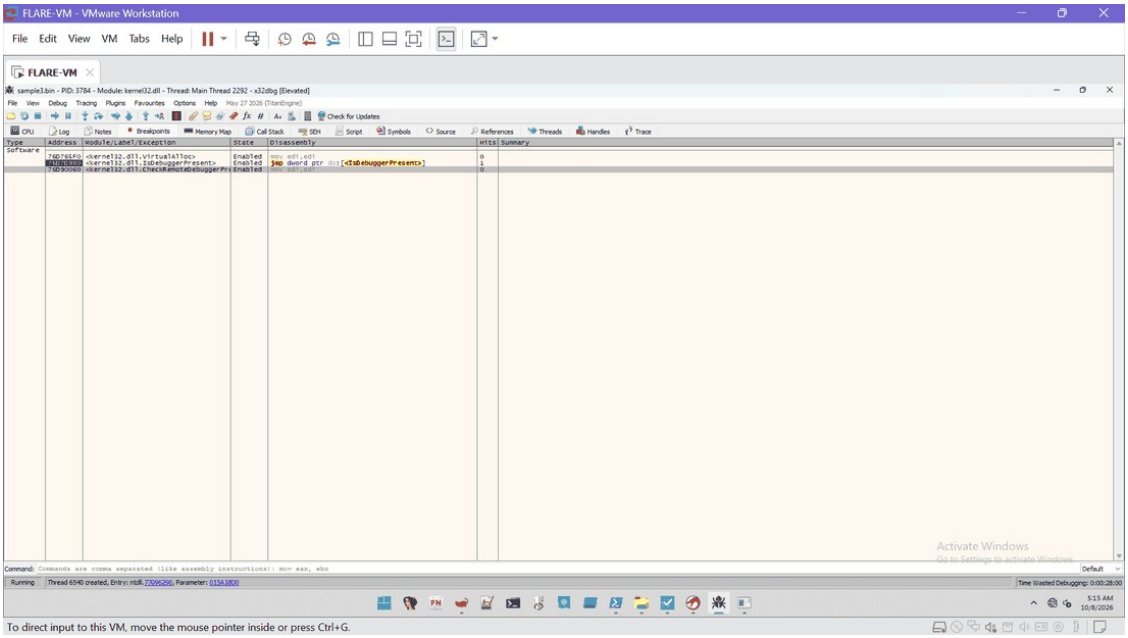

***Ex. E48*** | *2026-10-08 05:13* | *x32dbg* | *Run 3, PID 3784: TLS callback 1 (gdi32full) stop* | *SHA-256 C181B4542038A408…*



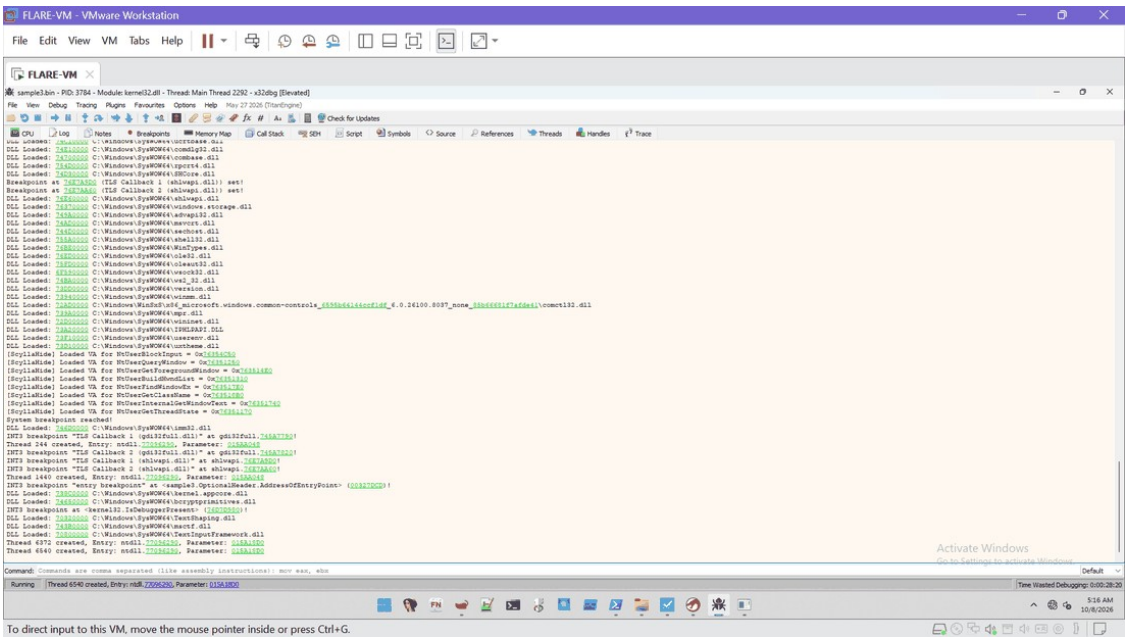

***Ex. E49*** | *2026-10-08 05:13* | *x32dbg* | *Run 3: INT3 breakpoint at kernel32.IsDebuggerPresent (76D7D980)* | *SHA-256 D2D1E4F06C5682B4…*




***Ex. E50*** | *2026-10-08 05:14 | x32dbg | Run 3 Call Stack at IsDebuggerPresent: return to sample3.00304864 (from 76D7D980), caller frames sample3.00304864 and sample3.00327D53 | SHA-256 8C250F0181B257E4…*



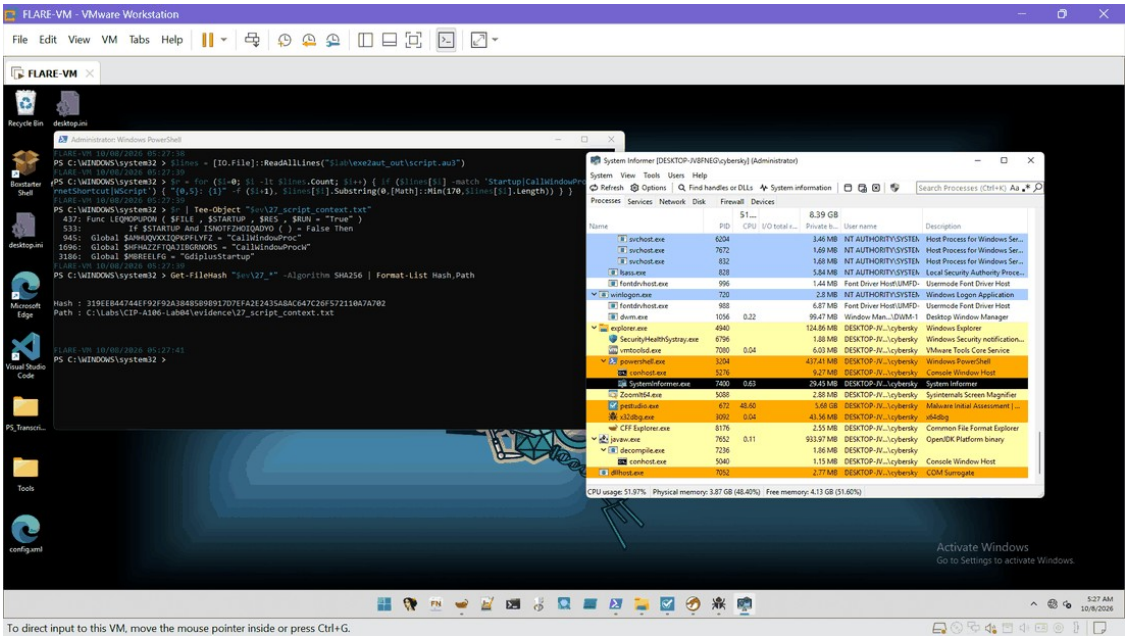

***Ex. E51*** | *2026-10-08 05:15 | x32dbg | Run 3 Breakpoints tab: VirtualAlloc 76D765F0 with 0 hits, IsDebuggerPresent hit count 1 | SHA-256 BAD8AA0748A0622F…*




***Ex. E52*** | *2026-10-08 05:16 | x32dbg | Run 3 log: ScyllaHide "Loaded VA" lines still printed with options unticked, TLS callbacks, entry breakpoint, IsDebuggerPresent breakpoint, TextInputFramework loaded | SHA-256 D06EA2AE156E5468…*

# E. Memory and script context (Tasks B and E)

System Informer and the PowerShell record of the script lines (27_script_context).




***Ex. E53*** | *2026-10-08 05:27 | System Informer / PowerShell | System Informer process tree (sample no longer running; x32dbg PID 3092, PowerShell 3204) beside PowerShell showing 27_script_context.txt: LEQMOPUPON function (line 437), $STARTUP test (line 533), CallWindowProc, CallWindowProcW and GdiplusStartup variables (lines 945, 1696, 3186), file SHA-256 319EEB44…A702 | SHA-256 AB46327BBC4CE5CA…*

# T. Rebuilt text exports

These files were copied out of the VM as console text (or transcribed from a screenshot where marked), because there is no clipboard. Line endings differ from the VM originals, so their hashes differ; the VM hash of each original is in the Evidence Manifest.

## T1. 25_indicator_counts.txt (18 patterns counted in script.au3 and the decoded strings, 05:25:52)

```
script.au3                  Charon              0
script.au3                  Inferno             0
script.au3                  Mozilla             0
script.au3                  fre.php             0
script.au3                  .php                0
script.au3                  Content-Key         0
script.au3                  MachineGuid         0
script.au3                  wininet             0
script.au3                  InternetOpen        0
script.au3                  HttpSendRequest     0
script.au3                  User-Agent          0
script.au3                  POST                0
script.au3                  Startup             3
script.au3                  .url                0
script.au3                  .vbs                0
script.au3                  KnownDlls           0
script.au3                  CallWindowProc      2
script.au3                  DllCall             0
30_au3_decoded_strings.txt Charon               0
30_au3_decoded_strings.txt Inferno              0
30_au3_decoded_strings.txt Mozilla              0
30_au3_decoded_strings.txt fre.php              0
30_au3_decoded_strings.txt .php                 0
30_au3_decoded_strings.txt Content-Key          0
30_au3_decoded_strings.txt MachineGuid          0
30_au3_decoded_strings.txt wininet              0
30_au3_decoded_strings.txt InternetOpen         0
30_au3_decoded_strings.txt HttpSendRequest      0
30_au3_decoded_strings.txt User-Agent           0
30_au3_decoded_strings.txt POST                 0
30_au3_decoded_strings.txt Startup              0
30_au3_decoded_strings.txt .url                 0
30_au3_decoded_strings.txt .vbs                 0
30_au3_decoded_strings.txt KnownDlls            0
30_au3_decoded_strings.txt CallWindowProc       0
30_au3_decoded_strings.txt DllCall              0
```

## T2. 26_after_run3_new_files.txt (files written after 05:00, run 3 check, 05:25:34)

```
PS > Get-ChildItem $env:USERPROFILE,$env:ProgramData,"C:\Users\Public","C:\Windows\Temp" -Recurse -File -Force -
Includ

FullName
--------
C:\Users\cybersky\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat
C:\Users\cybersky\AppData\Local\Packages\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\AC\Temp\1a07b68b-c481-4d15-aca3-
f..
C:\Users\cybersky\AppData\Roaming\Microsoft\Windows\Recent\sample3.bin.lnk
C:\Users\cybersky\AppData\Roaming\Microsoft\Windows\Recent\work.lnk

(26_after_run3_task_files.txt was not created: the System32\Tasks query returned no files newer than 05:00)

Get-FileHash result (console, 05:25:55):
772088FB4D38AD26805F60784635622F58C92A0248779B7D87BD4BCCB67021D7  25_indicator_counts.txt
47F745757DB87941F213336476584175817 31F69E364483FA612DA77B07B4FB2  26_after_run3_new_files.txt
```

## T3. 27_script_context.txt (script lines with Startup, CallWindowProc, GdiplusStartup, 05:27:39; transcribed from E53)

```
FLARE-VM 10/08/2026 05:27:38
PS C:\WINDOWS\system32 > $lines = [IO.File]::ReadAllLines("$lab\exe2aut_out\script.au3")
PS C:\WINDOWS\system32 > $r = for ($i=0; $i -lt $lines.Count; $i++) { if ($lines[$i] -match 'Startup|
CallWindowPro[cut
PS C:\WINDOWS\system32 > $r | Tee-Object "$ev\27_script_context.txt"
  437: Func LEQMOPUPON ( $FILE , $STARTUP , $RES , $RUN = "True" )
  533:          If $STARTUP And ISNOTFZHOIQADYO ( ) = False Then
  945:  Global $AMHUQVXXIQPKPFLYFZ = "CallWindowProc"
 1696:  Global $HFHAZZFTQAJIBGRNORS = "CallWindowProcW"
 3186:  Global $MBREELFG = "GdiplusStartup"
PS C:\WINDOWS\system32 > Get-FileHash "$ev\27_*" -Algorithm SHA256 | Format-List Hash,Path
Hash : 319EEB44744EF92F92A38485B98917D7EFA2E2435A8AC647C26F572110A7A702
Path : C:\Labs\CIP-A106-Lab04\evidence\27_script_context.txt
(Transcribed from the VM screenshot E53; the middle of the -match pattern is cut off on screen. Three lines contain
th
```

## T4. 30_au3_decoded_strings.txt (first lines of the decoded hex literals, 23:45:20; the full file stays in the VM)

```
FLARE-VM 10/07/2026 23:45:17
PS C:\WINDOWS\system32 > $lab = "C:\Labs\CIP-A106-Lab04"; $ev = "$lab\evidence"; $sc = "$lab\exe2aut_out\script.au3"
FLARE-VM 10/07/2026 23:45:19
... (hex-literal count and length table, see the VM transcript) ...
PS C:\WINDOWS\system32 > $m | ForEach-Object { $h=$_.Groups[2].Value; if ($h.Length % 2 -eq 0) { $b = for($i=0;$i -
lt
"[InternetShortcut]" & @CR & "URL=file:///" & StringReplace($hpudqgyrglybjztszujdupnyzjggzoyrledakhmgmngedexblxlu,
"\"
"Set WshShell = WScript.CreateObject(" & Chr(34) & "WScript.Shell" & Chr(34) & ")" & @CRLF & "WScript.Sleep(" &
Round(
$aagbrrpusetqgmijhtlv & "FFFF4E744372C78538FCFFFF65617465C7853CFCFFFF4D75746166C78540FCFFFF6E7488"
$aaimblpftybmllkrzjb & "C78554FDFFFF43727970C78558FDFFFF74446563C7855CFDFFFF72797074888560FDFFFF"
$afqiwxhbpqvgmqjanjpbumxzy & "C78514FCFFFF43727970C78518FCFFFF74446573C7851CFCFFFF74726F79C78520FCFFFF"
$aibrhlgprwbpuoyosluscjlnbchc & "BCEEFFFFE88CEBFFFF89450C8D85DCF6FFFFE87EEBFFFF89451485C0750E8D85ACEAFFFF"
$ampprdjio & "7D2981EC780100006A5E598BFCFF750C8D8424040300008DB4248C01000050F3A5E8CFFC"
$aqcvkvefl & "280345F8898540FCFFFF8D8590FBFFFF50FF75ECFF95600100005785C00F8831FFFFFFFF"
$ascikbidnjxeatwjmwyetkibgqnpdpmoeflzel = False
$asnzapvlzadkpbvs & "C78574FBFFFF436F4372C78578FBFFFF65617465C7857CFBFFFF496E7374C78580FBFFFF"
$avonnjn & "FF7357888592FCFFFFC78514FDFFFF52746C5AC78518FDFFFF65726F4DC7851CFDFFFF65"
$baewlspkneiucpinovcxaezpusywp & "8D8594FDFFFF508D85DCF6FFFF50FFD78D8510FEFFFF83C40C50FFD65940508D8510FEFF"
$bappvblh & "FF72656E74C785FCF8FFFF55736572C78500F9FFFF4B657950C78504F9FFFF61746800C7"
$bclbkflc & "8550FDFFFFC745BC6B65726EC745C0656C33328845C88845A8888580FDFFFF64A1300000"
... (remaining lines of the 80 shown are similar hex fragments)
FLARE-VM 10/07/2026 23:45:26
PS C:\WINDOWS\system32 >
```

## T5. Stack-string fragments read from the decoded hex literals (derived by the analyst, not tool output)

```
Derived by the analyst from the decoded hex literals in the paste (first 80 lines only).
The 4-byte ASCII fragments are read in order out of mov dword ptr [ebp-X],imm32 patterns
(opcode C7 85). The last fragment of a name can be cut by the line length, so the names
are reconstructions, not tool output. Nothing was disassembled or executed.

FFFF4E744372C785...  NtCr | eate | Mutaf
C78554FDFFFF4372...  Cryp | tDec | rypt
C78514FCFFFF4372...  Cryp | tDes | troy
C78574FBFFFF436F...  CoCr | eate | Inst
FF7357888592FCFF...  RtlZ | eroM
FF72656E74C785FC...  rent | User | KeyP | ath
8550FDFFFFC745BC...  kern | el32
6F6164C745B04C69...  oad | Libr | aryA | GetP
C745F070693332C7...  pi32 | .dll | lstr | len
74696F66C78520FA...  tiof | LdrG | etPr
C7858CFBFFFF7444...  tDes | troy | Hash
8001000085C0790C...  F4u
458C7763736366C7...  wcscf | memsf
856CF9FFFF6E7361...  nsac | tion | ZwRo
4B657900C78584FA...  Key | Cryp | tRel | ease
C785B8FAFFFF7254...  rThr | ead | NtOp
FFFF61746550C785...  ateP | rocef
6C735C75C78594FE...  ls\u | ser3 | 2.dlf
B0FBFFFF4E744465...  NtDe | layE | xecu
656E5072C78554FB...  enPr | oces | sTokf
FBFFFFC785ACF9FF...  NtAd | just | Priv
C78584FCFFFF4372...  Crea | tePr | ocesf
F9FFFF696C6567C7...  ileg | esTo | ken | Rtl
F9FFFF45787061C7...  Expa | ndEn | viro | nme
FFFF6579888562FC...  memcf | Cryp
57888538FFFFFFC7...  NtPr | otec | tVi
4B6E6FC78598FDFF...  Kno | wnDl | ls32 | \ker
32C78504FFFFFF2E...  .dll | user | 32.df
FF7300C78504FDFF...  GetS | yste | mInff
750C8BF8F3A55F5E...  SVW
FFFF74655769C785...  teWi | ndow | ExW | Regi
7233322EC78524FE...  r32. | dll | \Kno | wnDl
45C4774F6653C745...  wOfS | ectif
7CFAFFFF63657373...  cess | ZwCr | eate
6366C74598707988...  NtCr | eate
4765744DC78514FB...  GetM | odul | eFil | eNamf
D4FF958401000085...  PWWW
53565781EC780100...  SVW
6EC78528FCFFFF69...  itia | lizef
4F70C785E8FCFFFF...  enSe | ctiof
859CF9FFFF746556...  teVi | rtua | lMem
6164888582FCFFFF...  NtSe | tCon
DAFEFFFFC78534FD...  NtSe | tVal | ueKef
6D6F7266C78520FD...  morf | NtTe | rmin
8510FAFFFF6C6C62...  llba | ckTr | ansa
```

## T6. 21_after_run1_new_files.txt and the post-run-1 comparison (00:45 to 00:48)

```
FLARE-VM 10/08/2026 00:45:09
PS > Get-ChildItem $env:USERPROFILE,$env:ProgramData,"C:\Users\Public","C:\Windows\Temp" -Recurse -File -Force
     -Include *.vbs,*.url,*.lnk,*.js,*.bat,*.cmd,*.ps1,*.exe,*.dll,*.tmp,*.au3,*.dat
     | Where-Object LastWriteTime -gt (Get-Date "2026-10-08 00:19:00") | Sort-Object LastWriteTime

FullName                                                                        Length
--------                                                                        ------
C:\Users\cybersky\AppData\Roaming\Microsoft\Windows\Recent\sample3.bin.lnk          908
C:\Users\cybersky\AppData\Local\Microsoft\Windows\History\History.IE5\MSHist012026100820261009\container.dat   0
C:\Users\cybersky\AppData\Roaming\Microsoft\Windows\Recent\work.lnk                  718
C:\Users\cybersky\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat      14680064

FLARE-VM 10/08/2026 00:47:36  Run keys saved to 21_after1_run_keys.txt
FLARE-VM 10/08/2026 00:48:08  Scheduled tasks saved to 21_after1_tasks.txt
Compare-Object on run_keys : no differences
Compare-Object on tasks    : printed the whole task list (column wrapping made the text differ); judged inconclusive
(LastWriteTime column cut off in the console paste; the lines above are rebuilt from the pasted text)
```

## T7. Process and memory artefact record for run 3 (PID 3784, dump hash)

```
Values recorded for the run 3 process and the memory artefact. Rebuilt from console output noted during the lab;
the original console text is in the VM evidence folder (00_transcript.txt).

Process
  Name               sample3.bin
  ProcessId          3784
  Path               C:\Labs\CIP-A106-Lab04\work\sample3.bin
  Created            2026-10-08 05:12:12
  Parent ProcessId   3092 (x32dbg.exe, C:\Tools\x64dbg\release\x32\x32dbg.exe)

Memory artefact
  File               C:\Labs\CIP-A106-Lab04\memory\run3_blocked.dmp
  Written            2026-10-08 05:21:30
  Size               97,956,974 bytes
  SHA-256            3DFEF5111364B61EC8345DBC835964E1A88216D335B2114C7EBA026381EAD5E5
  Tool               System Informer 3.2.25113.1114, Create dump file (full dump)
  Validation         evidence\24_dump_validation.txt, SHA-256
D04E10B488B7F334ECB2C20C2EACE70E1202876FF5E1F5BC2ED26195
  Note               Format-Hex -Count is not supported in Windows PowerShell 5.1; bytes were read with
[IO.File]::Rea
```

## T8. SHA-256 values of the evidence files recorded in the VM

```
Evidence file hashes recorded in the VM (SHA-256, file name). Rebuilt from the manifest.

85B05D9BA00CD16DD370B628F8943D3C85EB87C23178CD30CCEECF00FB49974C  D:\sample3.iso
0DA91175E7D72A7FF2BCB3FD93F2BA7BBE4045F9C4DEE5C9685C7FDF6DA622A6  original\sample.bin
0DA91175E7D72A7FF2BCB3FD93F2BA7BBE4045F9C4DEE5C9685C7FDF6DA622A6  work\sample3.bin
788D259BE4F29DE5865FA18C4B0089F6A1F0D10C910376250AE6924F1E5B6099  exe2aut_out\script.au3
476D83A1B7835D9CA91ED156CEDE641BA3E1C74E627149A9F706C90EDF987DD6  evidence\20_baseline_files.csv
EE560871671E6361CFEA2ABC20CE1EA6D22E7CF6B87D8827C7144ADA60197C0A  evidence\20_baseline_processes.txt
92FDE6438637300E6BF7697941572765D573ACE69C03C83907AF6795A5D86E98  evidence\20_baseline_run_keys.txt
A746B5CD178FF855162B7EE0936DFD3CF735CDC5FD9C9AA8E007E7F640A29D44  evidence\20_baseline_tasks.txt
C0D62A7DE7B90D05814AE575136990A74E3394589646799244231E4B698DCE46  evidence\21_after_run1_time.txt
BDC3E8DDF892E45D2A2DB0DB38A9551F8828E23E2C77E04E5CC5FF9F392FB1F9  evidence\21_after_run1_new_files.txt
92FDE6438637300E6BF7697941572765D573ACE69C03C83907AF6795A5D86E98  evidence\21_after1_run_keys.txt
88B2112200BA6A698CDCC70306E12EF111EA2FCB449E3484C40439D4071258DB  evidence\21_after1_tasks.txt
674CEDF5EC162581252725FE9C4CBFD1EF71825EE0129B446F41C707D3EF4FF8  evidence\22_run1_x32dbg_db.dd32
952A9BAD1CFFC081D90DB925D60AD8E0933A4727DB87BB5048EEDCBC70B56B25  evidence\23_after_run2_new_files.txt
D551A91A2B99A0FE0A7A73D3BEF5BC348E49F7339A63764520755D722E4AB365  evidence\23_after_run2_task_files.txt
3DFEF5111364B61EC8345DBC835964E1A88216D335B2114C7EBA026381EAD5E5  memory\run3_blocked.dmp
D04E10B488B7F334ECB2C20C2EACE70E1202876FF5E1F5BC2ED26195014BD421  evidence\24_dump_validation.txt
47F745757DB87941F213336476584175817 31F69E364483FA612DA77B07B4FB2  evidence\26_after_run3_new_files.txt
772088FB4D38AD26805F60784635622F58C92A0248779B7D87BD4BCCB67021D7  evidence\25_indicator_counts.txt
319EEB44744EF92F92A38485B98917D7EFA2E2435A8AC647C26F572110A7A702  evidence\27_script_context.txt
```

## T9. 24_dump_validation.txt (validation of run3_blocked.dmp, output shown at 05:54:30)

```
FLARE-VM 10/08/2026 05:54:30
PS C:\WINDOWS\system32 > Get-Content "$ev\24_dump_validation.txt"
First 16 bytes: 4D 44 4D 50 93 A7 F4 A0 0F 00 00 00 20 00 00 00  ASCII: MDMP
This is a third-party compiled AutoIt script   ascii offset:    805858   utf16 offset:   2840286
AU3!EA06                                       ascii offset:    904246   utf16 offset:        -1
Mozilla/4.08                                   ascii offset:        -1   utf16 offset:        -1
Charon                                         ascii offset:        -1   utf16 offset:        -1
Inferno                                        ascii offset:        -1   utf16 offset:        -1
fre.php                                        ascii offset:        -1   utf16 offset:        -1
Content-Key                                    ascii offset:        -1   utf16 offset:        -1
Original file DOS header and stub (128 bytes) found in dump at offsets: 54382

FLARE-VM 10/08/2026 05:54:30
PS C:\WINDOWS\system32 > Get-FileHash "$ev\24_dump_validation.txt" -Algorithm SHA256
SHA256  D04E10B488B7F334ECB2C20C2EACE70E1202876FF5E1F5BC2ED26195014BD421  C:\Labs\CIP-A106-Lab04\eviden...

(Offset -1 means the pattern was not found. Offsets are byte positions in run3_blocked.dmp. Rebuilt from the console
t
```